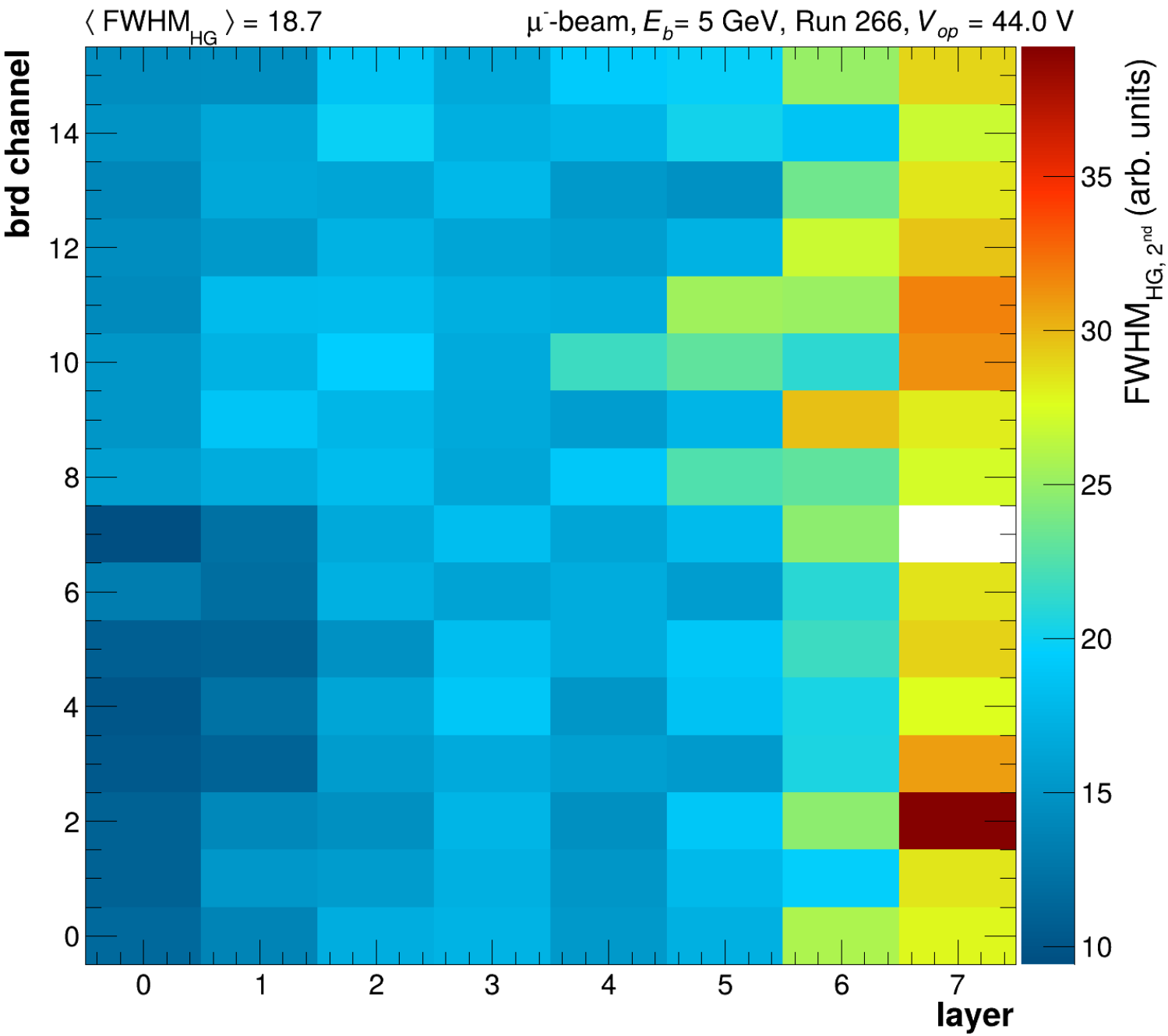
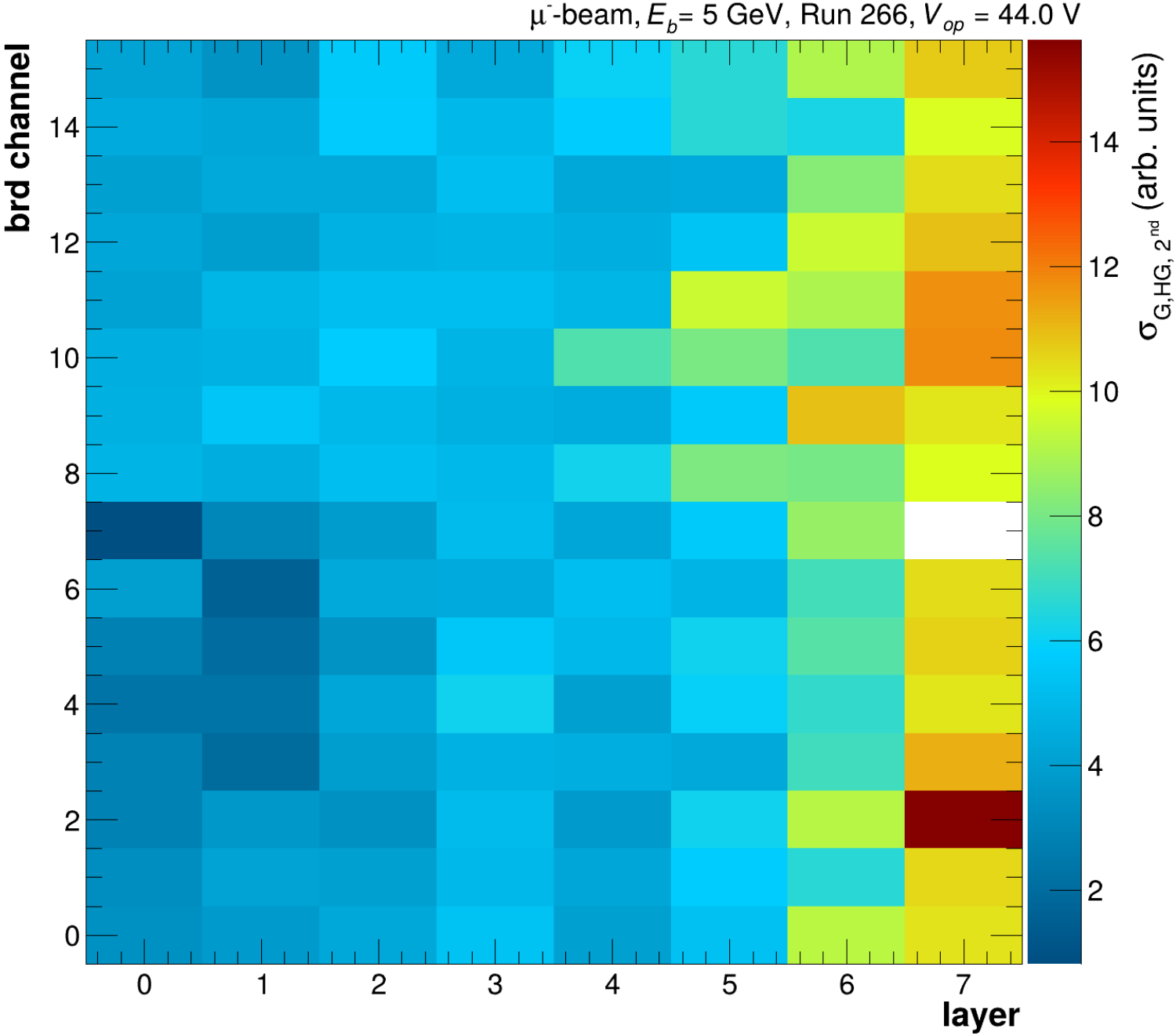
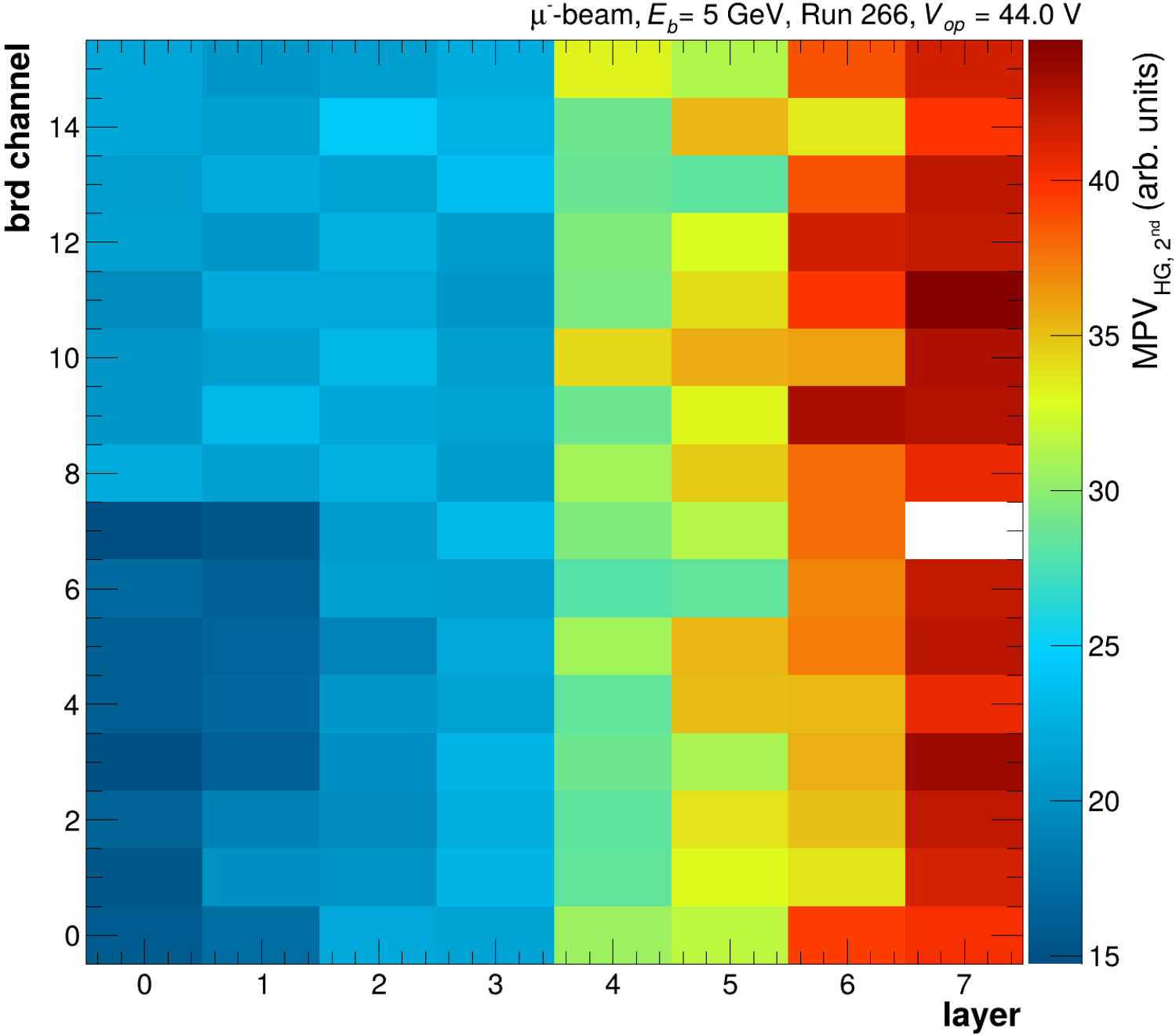
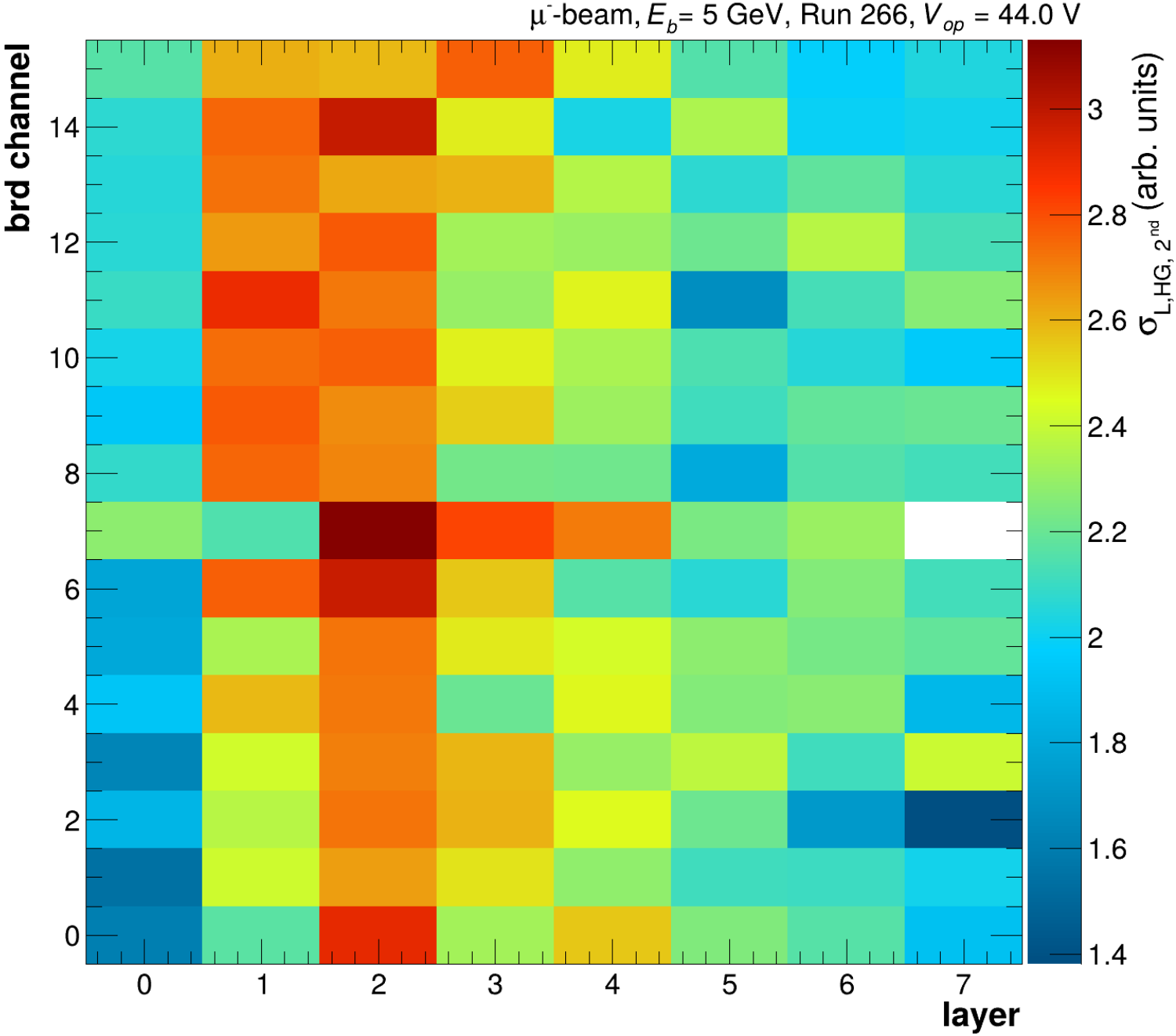
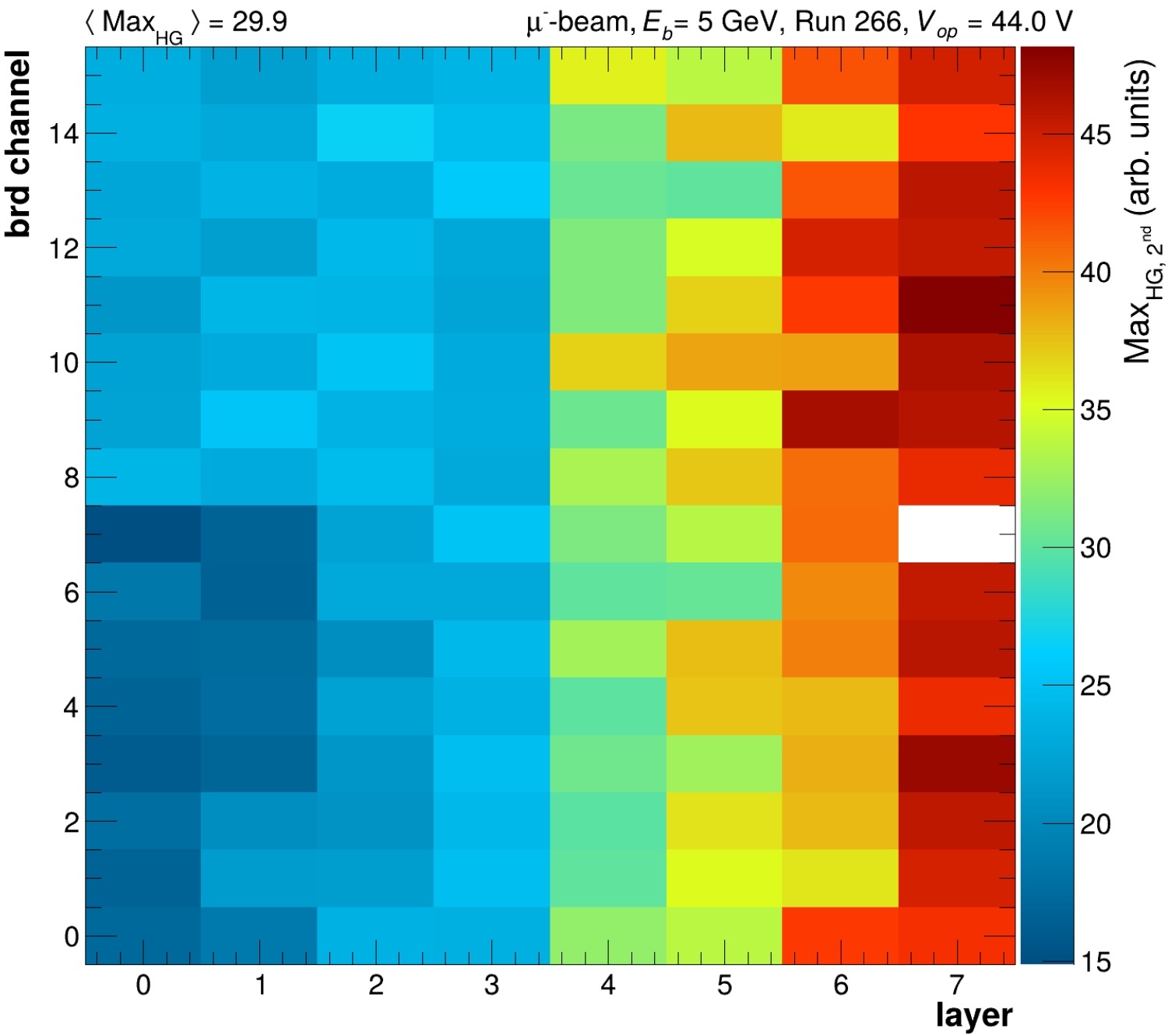
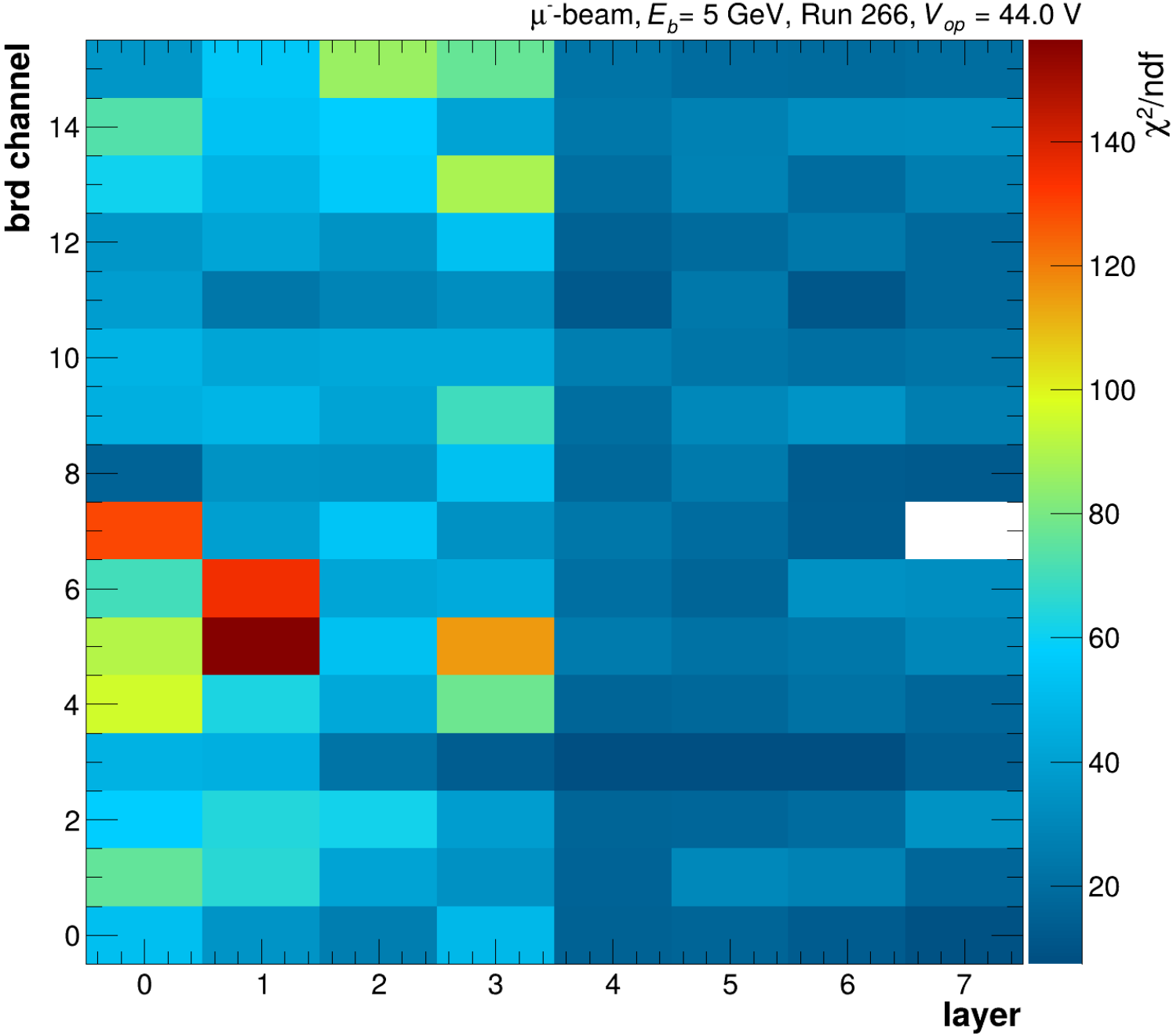

⟨ FWHM$_{HG}$ ⟩ = 18.7
μ$^-$-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
FWHM$_{HG,\ 2^{nd}}$ (arb. units)
10
15
20
25
30
35

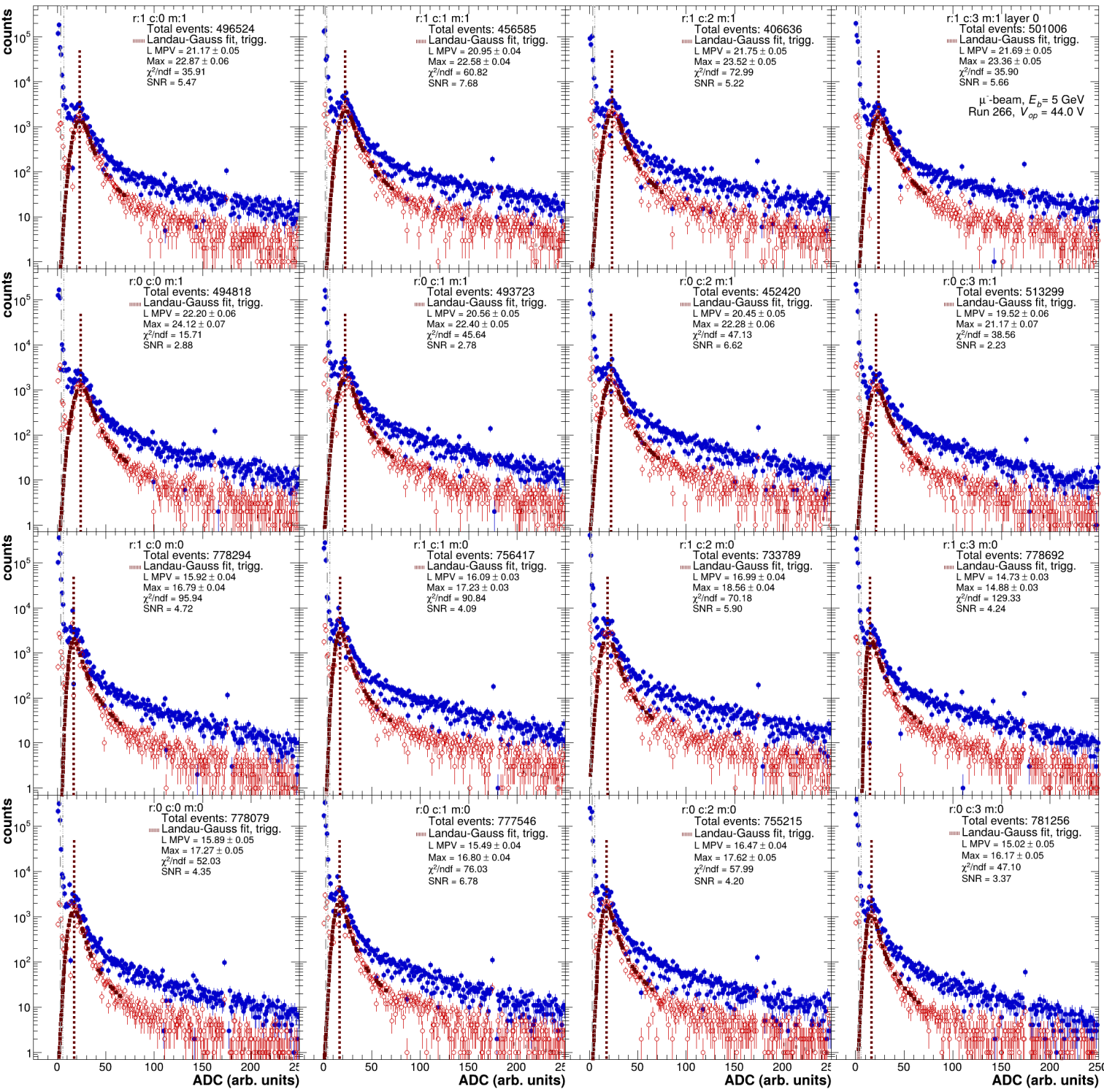

μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

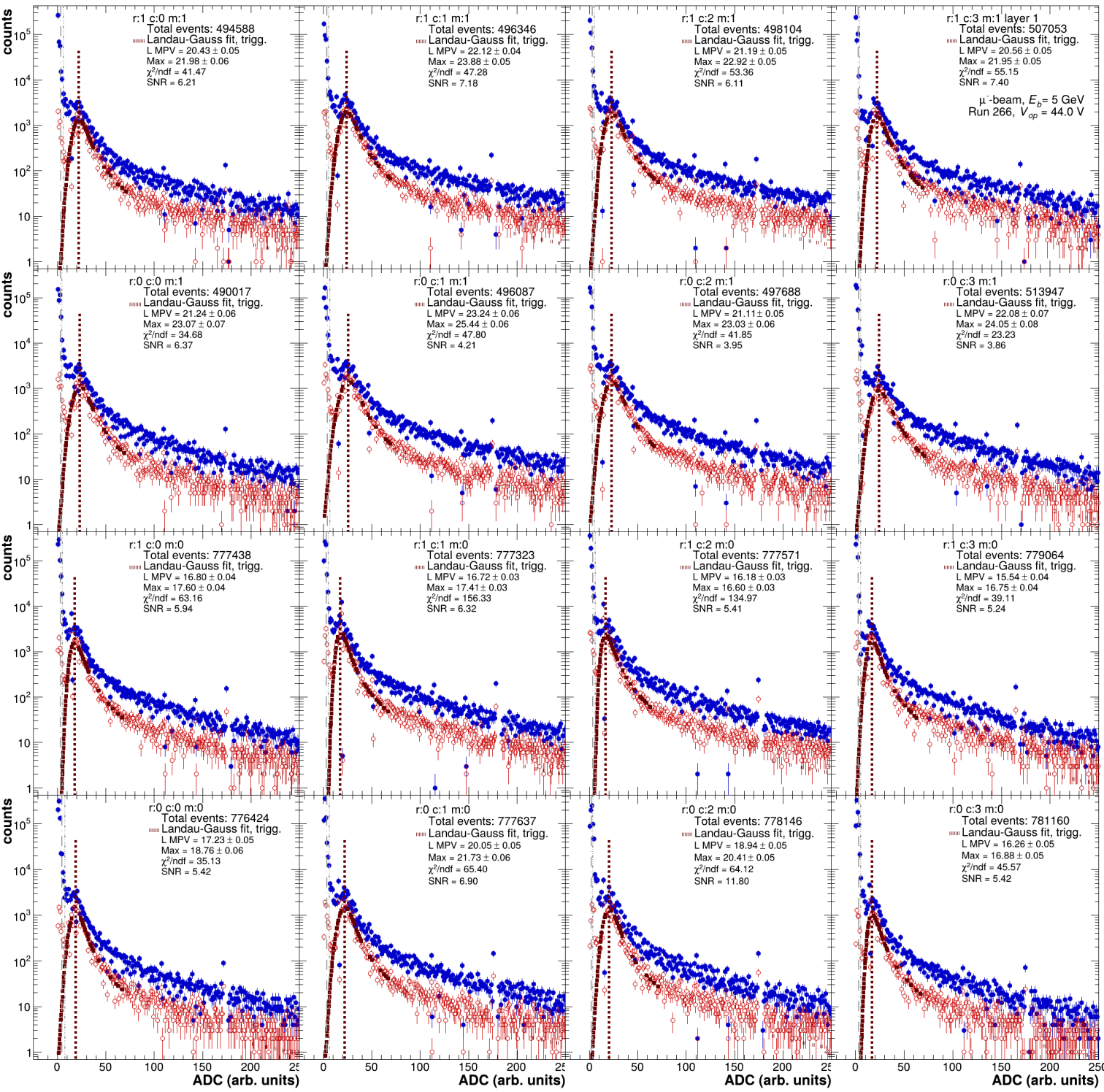

μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$MPV_{HG,\ 2^{nd}}$ (arb. units)
15
20
25
30
35
40

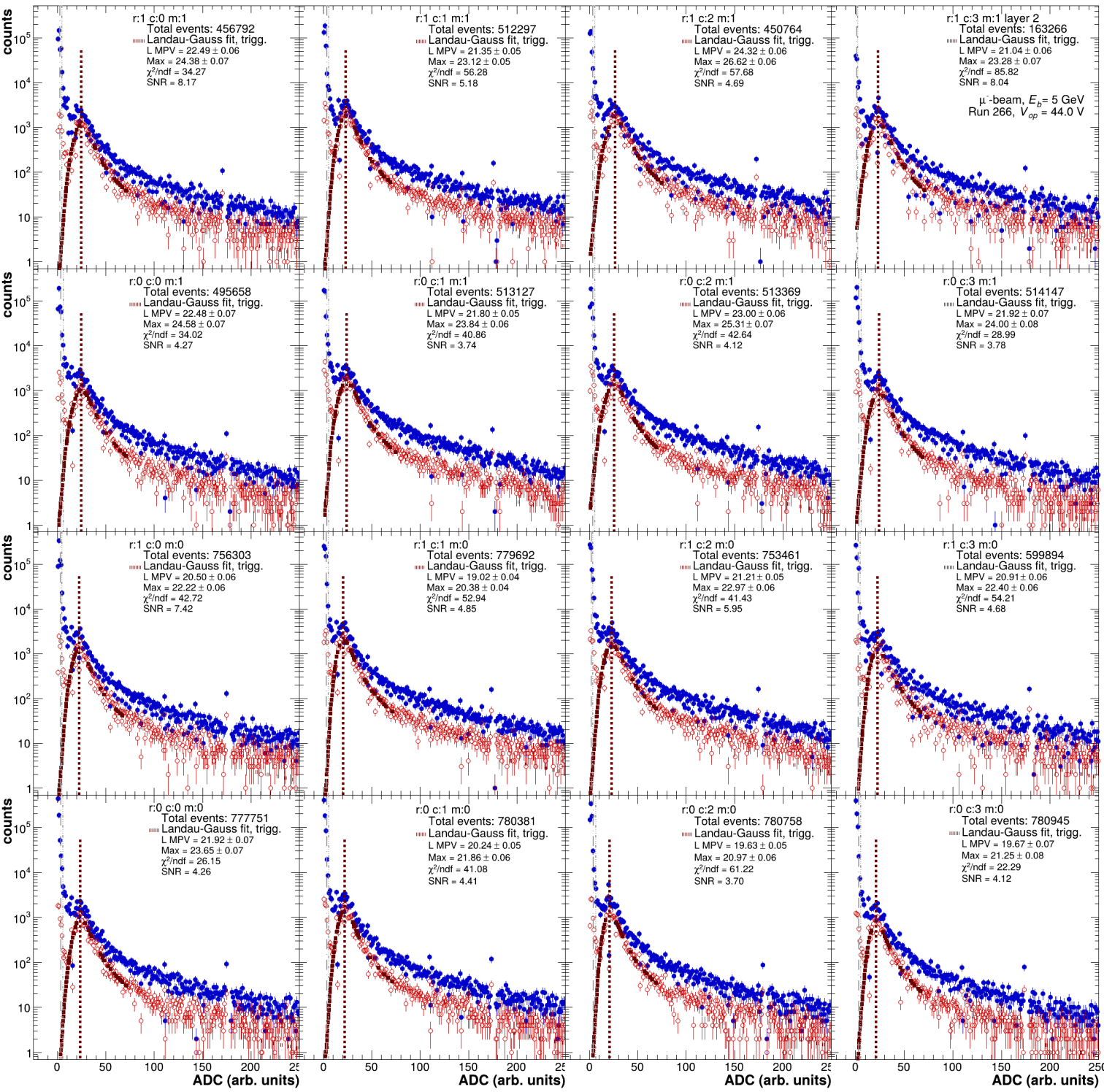

μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
3
2.8
2.6
2.4
2.2
2
1.8
1.6
1.4

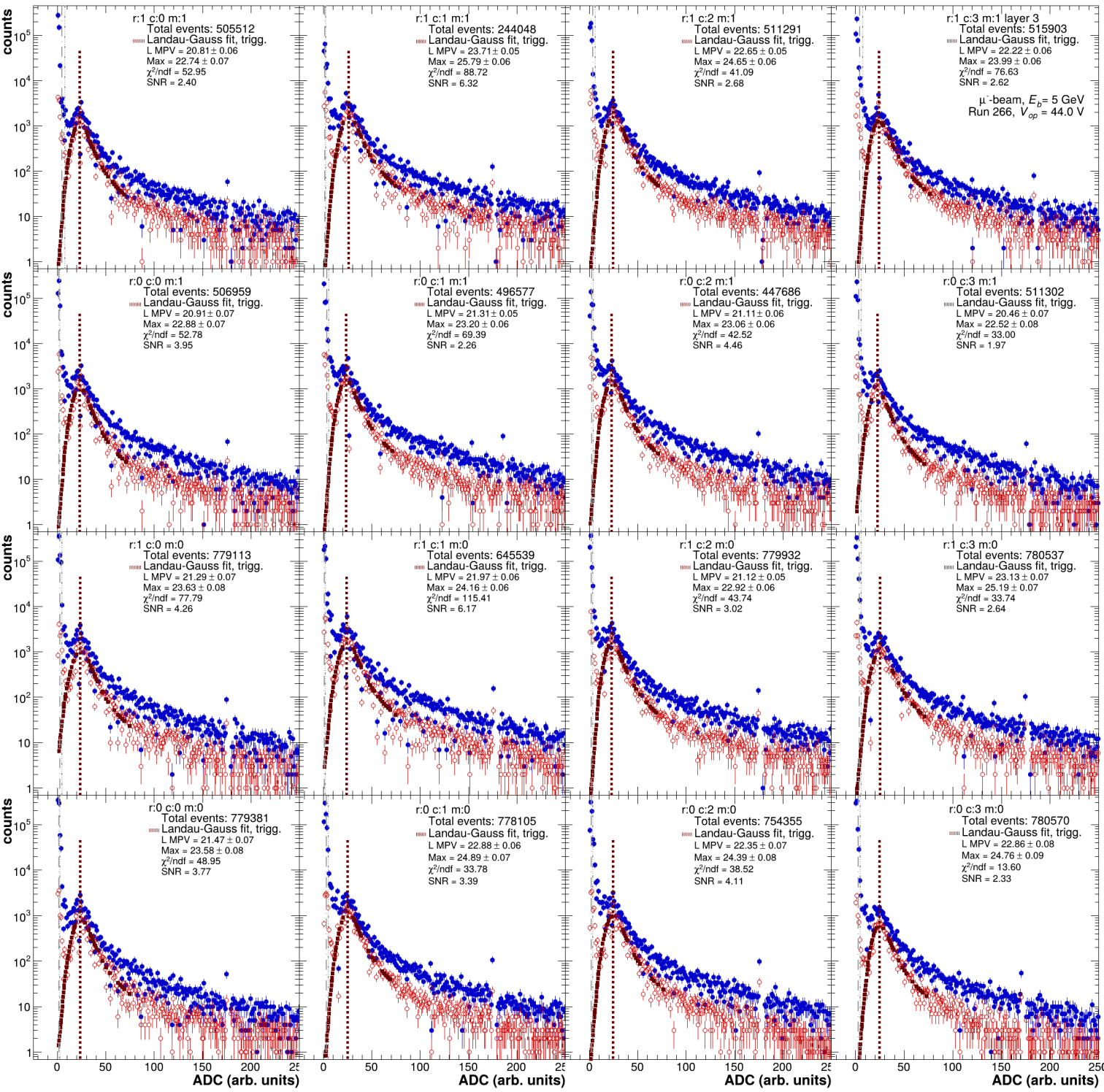

⟨ Max_HG ⟩ = 29.9
μ⁻-beam, E_b = 5 GeV, Run 266, V_op = 44.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
15
20
25
30
35
40
45

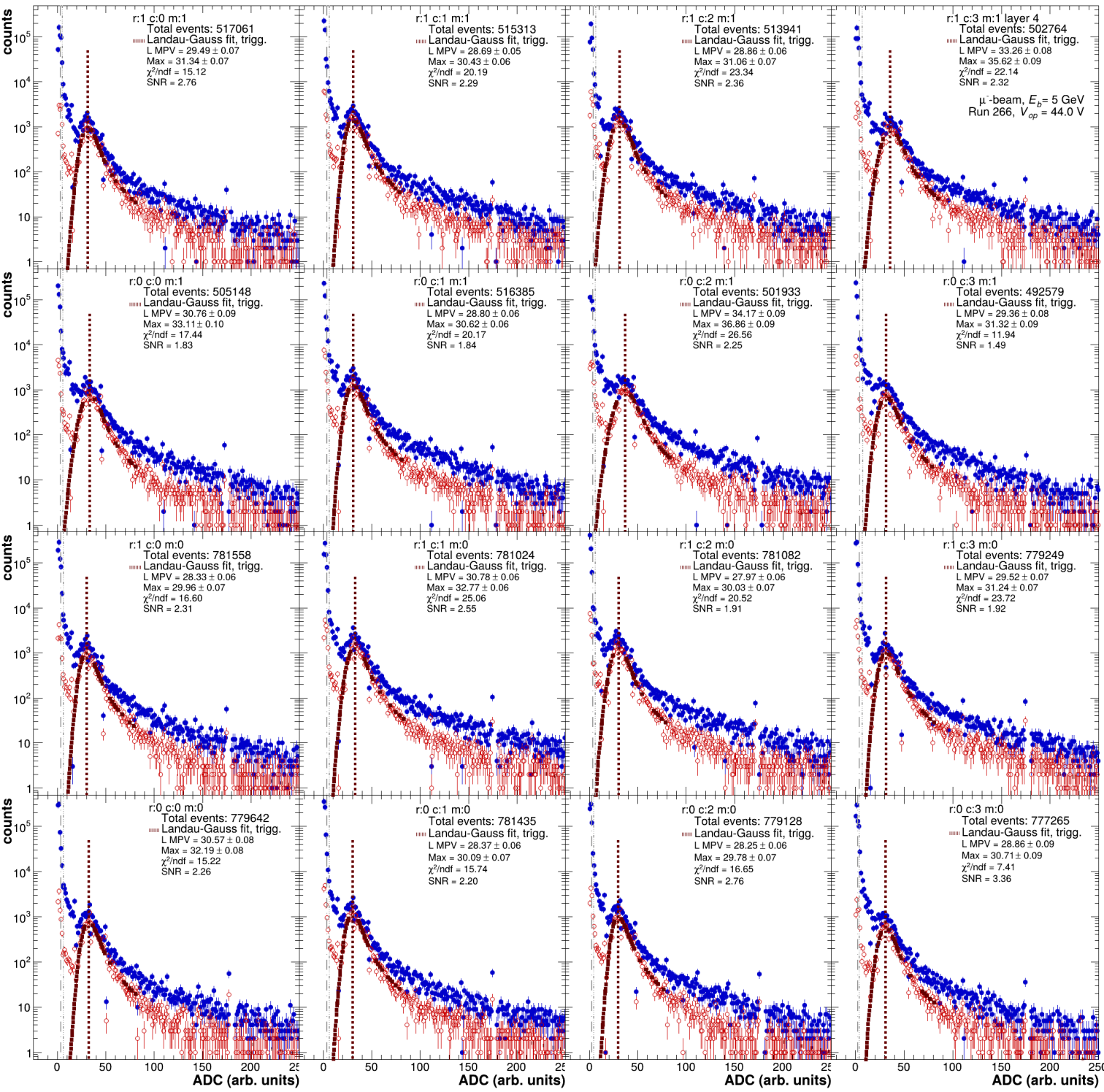

μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
20
40
60
80
100
120
140

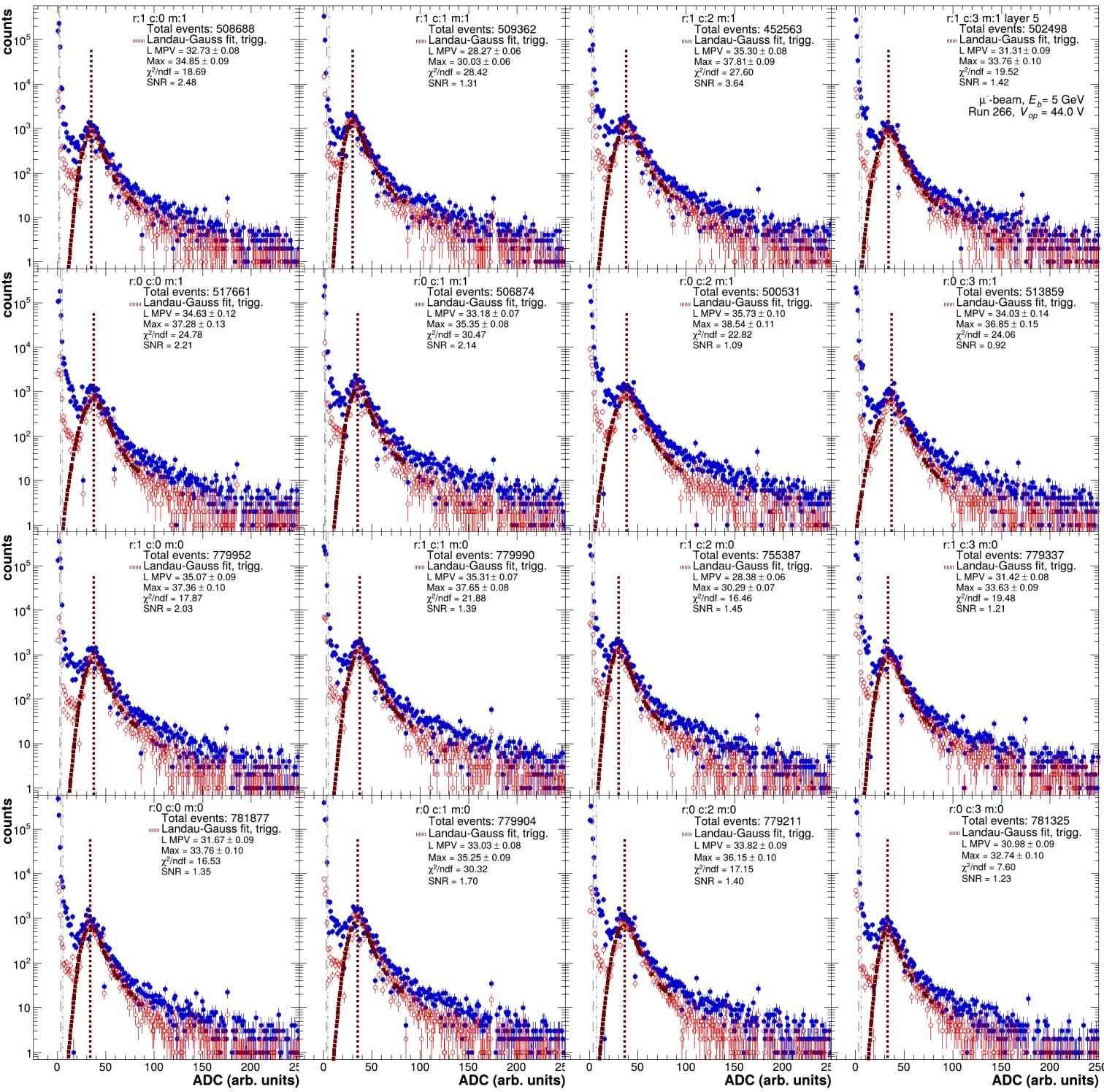

r:1 c:0 m:1
Total events: 496524
Landau-Gauss fit, trigg.
L MPV = 21.17 ± 0.05
Max = 22.87 ± 0.06
χ²/ndf = 35.91
SNR = 5.47
r:1 c:1 m:1
Total events: 456585
Landau-Gauss fit, trigg.
L MPV = 20.95 ± 0.04
Max = 22.58 ± 0.04
χ²/ndf = 60.82
SNR = 7.68
r:1 c:2 m:1
Total events: 406636
Landau-Gauss fit, trigg.
L MPV = 21.75 ± 0.05
Max = 23.52 ± 0.05
χ²/ndf = 72.99
SNR = 5.22
r:1 c:3 m:1 layer 0
Total events: 501006
Landau-Gauss fit, trigg.
L MPV = 21.69 ± 0.05
Max = 23.36 ± 0.05
χ²/ndf = 35.90
SNR = 5.66
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 494818
Landau-Gauss fit, trigg.
L MPV = 22.20 ± 0.06
Max = 24.12 ± 0.07
χ²/ndf = 15.71
SNR = 2.88
r:0 c:1 m:1
Total events: 493723
Landau-Gauss fit, trigg.
L MPV = 20.56 ± 0.05
Max = 22.40 ± 0.05
χ²/ndf = 45.64
SNR = 2.78
r:0 c:2 m:1
Total events: 452420
Landau-Gauss fit, trigg.
L MPV = 20.45 ± 0.05
Max = 22.28 ± 0.06
χ²/ndf = 47.13
SNR = 6.62
r:0 c:3 m:1
Total events: 513299
Landau-Gauss fit, trigg.
L MPV = 19.52 ± 0.06
Max = 21.17 ± 0.07
χ²/ndf = 38.56
SNR = 2.23
r:1 c:0 m:0
Total events: 778294
Landau-Gauss fit, trigg.
L MPV = 15.92 ± 0.04
Max = 16.79 ± 0.04
χ²/ndf = 95.94
SNR = 4.72
r:1 c:1 m:0
Total events: 756417
Landau-Gauss fit, trigg.
L MPV = 16.09 ± 0.03
Max = 17.23 ± 0.03
χ²/ndf = 90.84
SNR = 4.09
r:1 c:2 m:0
Total events: 733789
Landau-Gauss fit, trigg.
L MPV = 16.99 ± 0.04
Max = 18.56 ± 0.04
χ²/ndf = 70.18
SNR = 5.90
r:1 c:3 m:0
Total events: 778692
Landau-Gauss fit, trigg.
L MPV = 14.73 ± 0.03
Max = 14.88 ± 0.03
χ²/ndf = 129.33
SNR = 4.24
r:0 c:0 m:0
Total events: 778079
Landau-Gauss fit, trigg.
L MPV = 15.89 ± 0.05
Max = 17.27 ± 0.05
χ²/ndf = 52.03
SNR = 4.35
r:0 c:1 m:0
Total events: 777546
Landau-Gauss fit, trigg.
L MPV = 15.49 ± 0.04
Max = 16.80 ± 0.04
χ²/ndf = 76.03
SNR = 6.78
r:0 c:2 m:0
Total events: 755215
Landau-Gauss fit, trigg.
L MPV = 16.47 ± 0.04
Max = 17.62 ± 0.05
χ²/ndf = 57.99
SNR = 4.20
r:0 c:3 m:0
Total events: 781256
Landau-Gauss fit, trigg.
L MPV = 15.02 ± 0.05
Max = 16.17 ± 0.05
χ²/ndf = 47.10
SNR = 3.37
counts
ADC (arb. units)

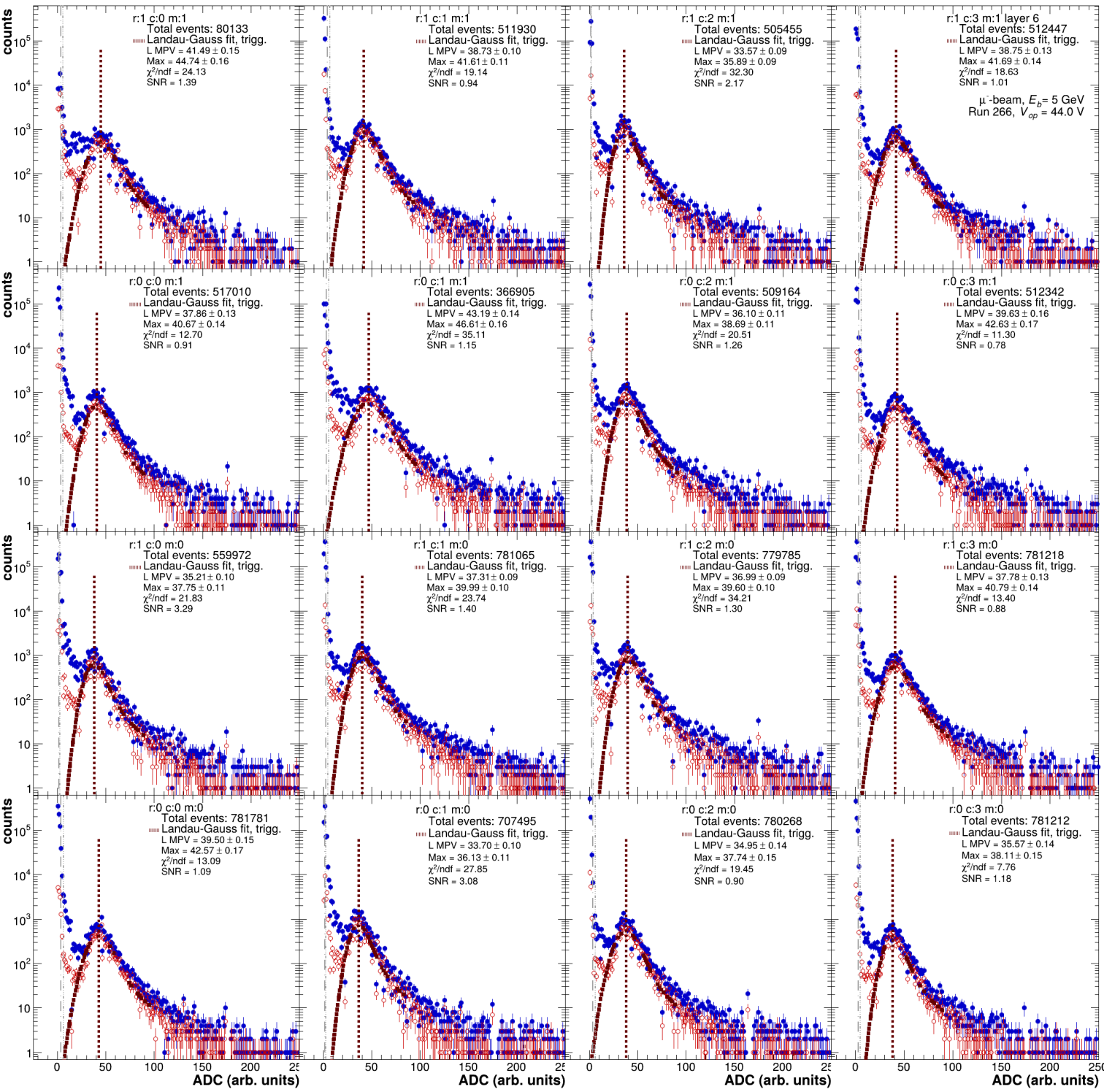

r:1 c:0 m:1
Total events: 494588
Landau-Gauss fit, trigg.
L MPV = 20.43 ± 0.05
Max = 21.98 ± 0.06
χ²/ndf = 41.47
SNR = 6.21
r:1 c:1 m:1
Total events: 496346
Landau-Gauss fit, trigg.
L MPV = 22.12 ± 0.04
Max = 23.88 ± 0.05
χ²/ndf = 47.28
SNR = 7.18
r:1 c:2 m:1
Total events: 498104
Landau-Gauss fit, trigg.
L MPV = 21.19 ± 0.05
Max = 22.92 ± 0.05
χ²/ndf = 53.36
SNR = 6.11
r:1 c:3 m:1 layer 1
Total events: 507053
Landau-Gauss fit, trigg.
L MPV = 20.56 ± 0.05
Max = 21.95 ± 0.05
χ²/ndf = 55.15
SNR = 7.40
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 490017
Landau-Gauss fit, trigg.
L MPV = 21.24 ± 0.06
Max = 23.07 ± 0.07
χ²/ndf = 34.68
SNR = 6.37
r:0 c:1 m:1
Total events: 496087
Landau-Gauss fit, trigg.
L MPV = 23.24 ± 0.06
Max = 25.44 ± 0.06
χ²/ndf = 47.80
SNR = 4.21
r:0 c:2 m:1
Total events: 497688
Landau-Gauss fit, trigg.
L MPV = 21.11 ± 0.05
Max = 23.03 ± 0.06
χ²/ndf = 41.85
SNR = 3.95
r:0 c:3 m:1
Total events: 513947
Landau-Gauss fit, trigg.
L MPV = 22.08 ± 0.07
Max = 24.05 ± 0.08
χ²/ndf = 23.23
SNR = 3.86
r:1 c:0 m:0
Total events: 777438
Landau-Gauss fit, trigg.
L MPV = 16.80 ± 0.04
Max = 17.60 ± 0.04
χ²/ndf = 63.16
SNR = 5.94
r:1 c:1 m:0
Total events: 777323
Landau-Gauss fit, trigg.
L MPV = 16.72 ± 0.03
Max = 17.41 ± 0.03
χ²/ndf = 156.33
SNR = 6.32
r:1 c:2 m:0
Total events: 777571
Landau-Gauss fit, trigg.
L MPV = 16.18 ± 0.03
Max = 16.60 ± 0.03
χ²/ndf = 134.97
SNR = 5.41
r:1 c:3 m:0
Total events: 779064
Landau-Gauss fit, trigg.
L MPV = 15.54 ± 0.04
Max = 16.75 ± 0.04
χ²/ndf = 39.11
SNR = 5.24
r:0 c:0 m:0
Total events: 776424
Landau-Gauss fit, trigg.
L MPV = 17.23 ± 0.05
Max = 18.76 ± 0.06
χ²/ndf = 35.13
SNR = 5.42
r:0 c:1 m:0
Total events: 777637
Landau-Gauss fit, trigg.
L MPV = 20.05 ± 0.05
Max = 21.73 ± 0.06
χ²/ndf = 65.40
SNR = 6.90
r:0 c:2 m:0
Total events: 778146
Landau-Gauss fit, trigg.
L MPV = 18.94 ± 0.05
Max = 20.41 ± 0.05
χ²/ndf = 64.12
SNR = 11.80
r:0 c:3 m:0
Total events: 781160
Landau-Gauss fit, trigg.
L MPV = 16.26 ± 0.05
Max = 16.88 ± 0.05
χ²/ndf = 45.57
SNR = 5.42
counts
ADC (arb. units)

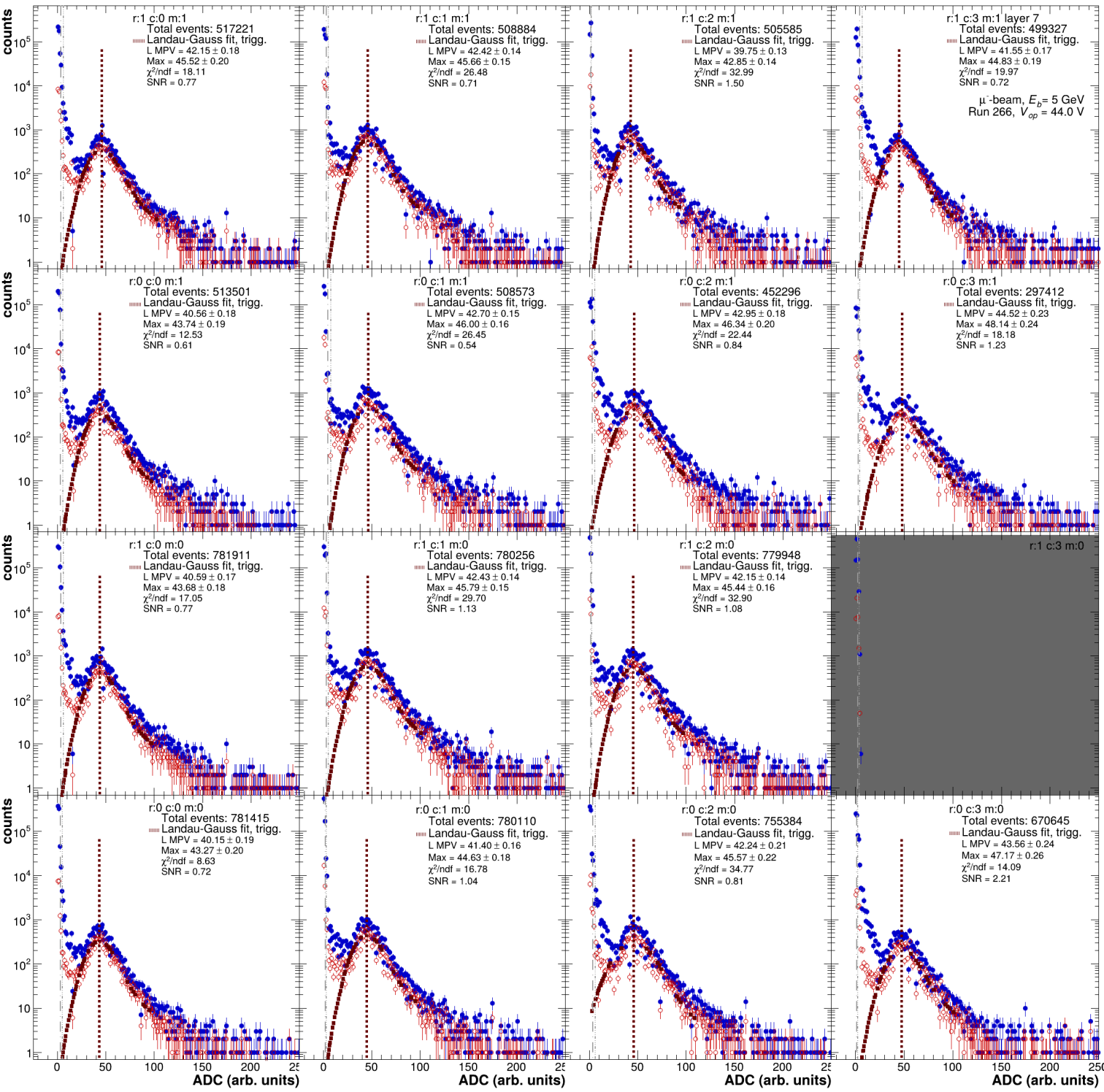

r:1 c:0 m:1
Total events: 456792
Landau-Gauss fit, trigg.
L MPV = 22.49 ± 0.06
Max = 24.38 ± 0.07
χ²/ndf = 34.27
SNR = 8.17
r:1 c:1 m:1
Total events: 512297
Landau-Gauss fit, trigg.
L MPV = 21.35 ± 0.05
Max = 23.12 ± 0.05
χ²/ndf = 56.28
SNR = 5.18
r:1 c:2 m:1
Total events: 450764
Landau-Gauss fit, trigg.
L MPV = 24.32 ± 0.06
Max = 26.62 ± 0.06
χ²/ndf = 57.68
SNR = 4.69
r:1 c:3 m:1 layer 2
Total events: 163266
Landau-Gauss fit, trigg.
L MPV = 21.04 ± 0.06
Max = 23.28 ± 0.07
χ²/ndf = 85.82
SNR = 8.04
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 495658
Landau-Gauss fit, trigg.
L MPV = 22.48 ± 0.07
Max = 24.58 ± 0.07
χ²/ndf = 34.02
SNR = 4.27
r:0 c:1 m:1
Total events: 513127
Landau-Gauss fit, trigg.
L MPV = 21.80 ± 0.05
Max = 23.84 ± 0.06
χ²/ndf = 40.86
SNR = 3.74
r:0 c:2 m:1
Total events: 513369
Landau-Gauss fit, trigg.
L MPV = 23.00 ± 0.06
Max = 25.31 ± 0.07
χ²/ndf = 42.64
SNR = 4.12
r:0 c:3 m:1
Total events: 514147
Landau-Gauss fit, trigg.
L MPV = 21.92 ± 0.07
Max = 24.00 ± 0.08
χ²/ndf = 28.99
SNR = 3.78
r:1 c:0 m:0
Total events: 756303
Landau-Gauss fit, trigg.
L MPV = 20.50 ± 0.06
Max = 22.22 ± 0.06
χ²/ndf = 42.72
SNR = 7.42
r:1 c:1 m:0
Total events: 779692
Landau-Gauss fit, trigg.
L MPV = 19.02 ± 0.04
Max = 20.38 ± 0.04
χ²/ndf = 52.94
SNR = 4.85
r:1 c:2 m:0
Total events: 753461
Landau-Gauss fit, trigg.
L MPV = 21.21 ± 0.05
Max = 22.97 ± 0.06
χ²/ndf = 41.43
SNR = 5.95
r:1 c:3 m:0
Total events: 599894
Landau-Gauss fit, trigg.
L MPV = 20.91 ± 0.06
Max = 22.40 ± 0.06
χ²/ndf = 54.21
SNR = 4.68
r:0 c:0 m:0
Total events: 777751
Landau-Gauss fit, trigg.
L MPV = 21.92 ± 0.07
Max = 23.65 ± 0.07
χ²/ndf = 26.15
SNR = 4.26
r:0 c:1 m:0
Total events: 780381
Landau-Gauss fit, trigg.
L MPV = 20.24 ± 0.05
Max = 21.86 ± 0.06
χ²/ndf = 41.08
SNR = 4.41
r:0 c:2 m:0
Total events: 780758
Landau-Gauss fit, trigg.
L MPV = 19.63 ± 0.05
Max = 20.97 ± 0.06
χ²/ndf = 61.22
SNR = 3.70
r:0 c:3 m:0
Total events: 780945
Landau-Gauss fit, trigg.
L MPV = 19.67 ± 0.07
Max = 21.25 ± 0.08
χ²/ndf = 22.29
SNR = 4.12
counts
ADC (arb. units)

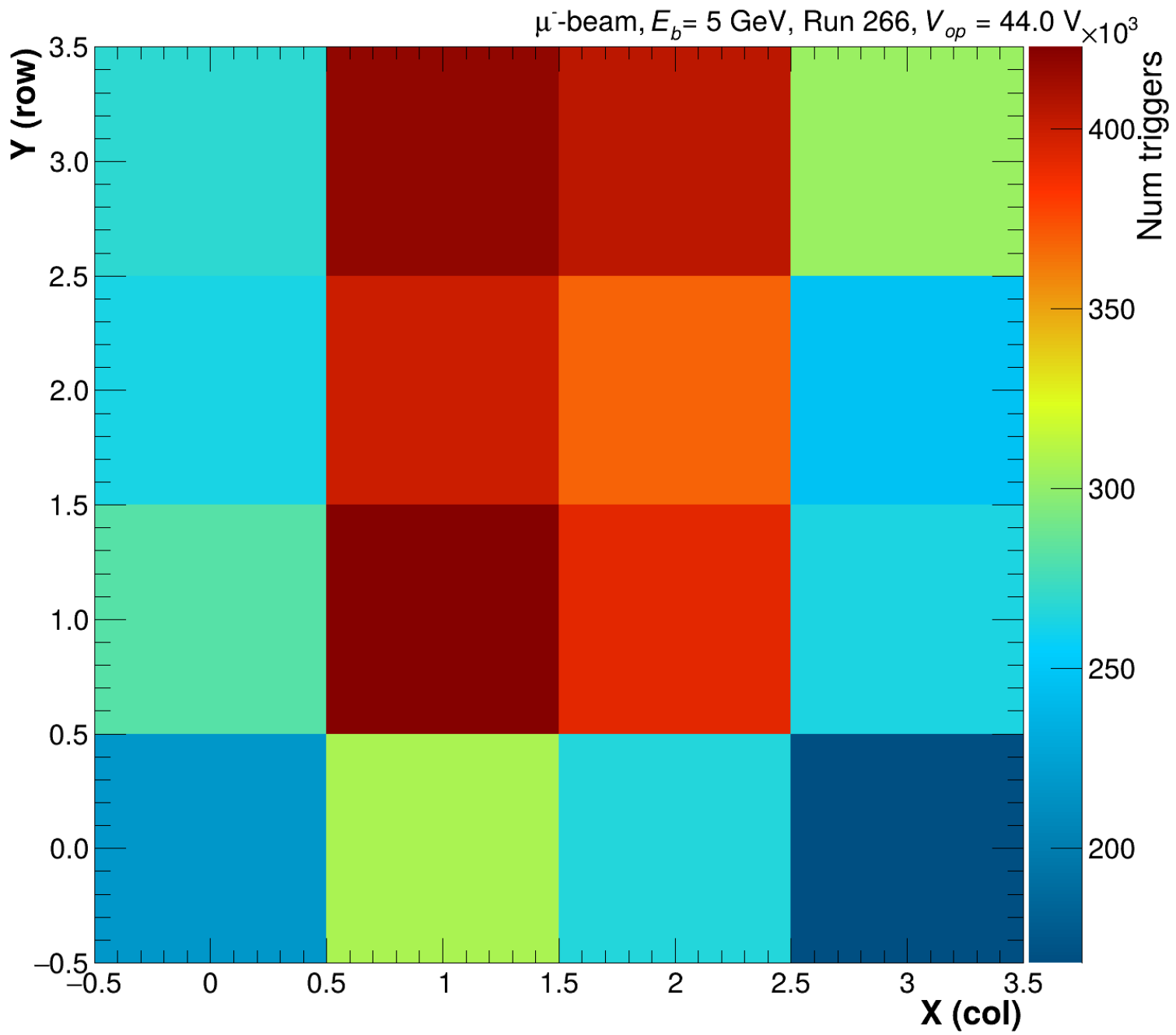

r:1 c:0 m:1
Total events: 505512
Landau-Gauss fit, trigg.
L MPV = 20.81 ± 0.06
Max = 22.74 ± 0.07
χ²/ndf = 52.95
SNR = 2.40
r:1 c:1 m:1
Total events: 244048
Landau-Gauss fit, trigg.
L MPV = 23.71 ± 0.05
Max = 25.79 ± 0.06
χ²/ndf = 88.72
SNR = 6.32
r:1 c:2 m:1
Total events: 511291
Landau-Gauss fit, trigg.
L MPV = 22.65 ± 0.05
Max = 24.65 ± 0.06
χ²/ndf = 41.09
SNR = 2.68
r:1 c:3 m:1 layer 3
Total events: 515903
Landau-Gauss fit, trigg.
L MPV = 22.22 ± 0.06
Max = 23.99 ± 0.06
χ²/ndf = 76.63
SNR = 2.62
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 506959
Landau-Gauss fit, trigg.
L MPV = 20.91 ± 0.07
Max = 22.88 ± 0.07
χ²/ndf = 52.78
SNR = 3.95
r:0 c:1 m:1
Total events: 496577
Landau-Gauss fit, trigg.
L MPV = 21.31 ± 0.05
Max = 23.20 ± 0.06
χ²/ndf = 69.39
SNR = 2.26
r:0 c:2 m:1
Total events: 447686
Landau-Gauss fit, trigg.
L MPV = 21.11 ± 0.06
Max = 23.06 ± 0.06
χ²/ndf = 42.52
SNR = 4.46
r:0 c:3 m:1
Total events: 511302
Landau-Gauss fit, trigg.
L MPV = 20.46 ± 0.07
Max = 22.52 ± 0.08
χ²/ndf = 33.00
SNR = 1.97
r:1 c:0 m:0
Total events: 779113
Landau-Gauss fit, trigg.
L MPV = 21.29 ± 0.07
Max = 23.63 ± 0.08
χ²/ndf = 77.79
SNR = 4.26
r:1 c:1 m:0
Total events: 645539
Landau-Gauss fit, trigg.
L MPV = 21.97 ± 0.06
Max = 24.16 ± 0.06
χ²/ndf = 115.41
SNR = 6.17
r:1 c:2 m:0
Total events: 779932
Landau-Gauss fit, trigg.
L MPV = 21.12 ± 0.05
Max = 22.92 ± 0.06
χ²/ndf = 43.74
SNR = 3.02
r:1 c:3 m:0
Total events: 780537
Landau-Gauss fit, trigg.
L MPV = 23.13 ± 0.07
Max = 25.19 ± 0.07
χ²/ndf = 33.74
SNR = 2.64
r:0 c:0 m:0
Total events: 779381
Landau-Gauss fit, trigg.
L MPV = 21.47 ± 0.07
Max = 23.58 ± 0.08
χ²/ndf = 48.95
SNR = 3.77
r:0 c:1 m:0
Total events: 778105
Landau-Gauss fit, trigg.
L MPV = 22.88 ± 0.06
Max = 24.89 ± 0.07
χ²/ndf = 33.78
SNR = 3.39
r:0 c:2 m:0
Total events: 754355
Landau-Gauss fit, trigg.
L MPV = 22.35 ± 0.07
Max = 24.39 ± 0.08
χ²/ndf = 38.52
SNR = 4.11
r:0 c:3 m:0
Total events: 780570
Landau-Gauss fit, trigg.
L MPV = 22.86 ± 0.08
Max = 24.76 ± 0.09
χ²/ndf = 13.60
SNR = 2.33
counts
ADC (arb. units)

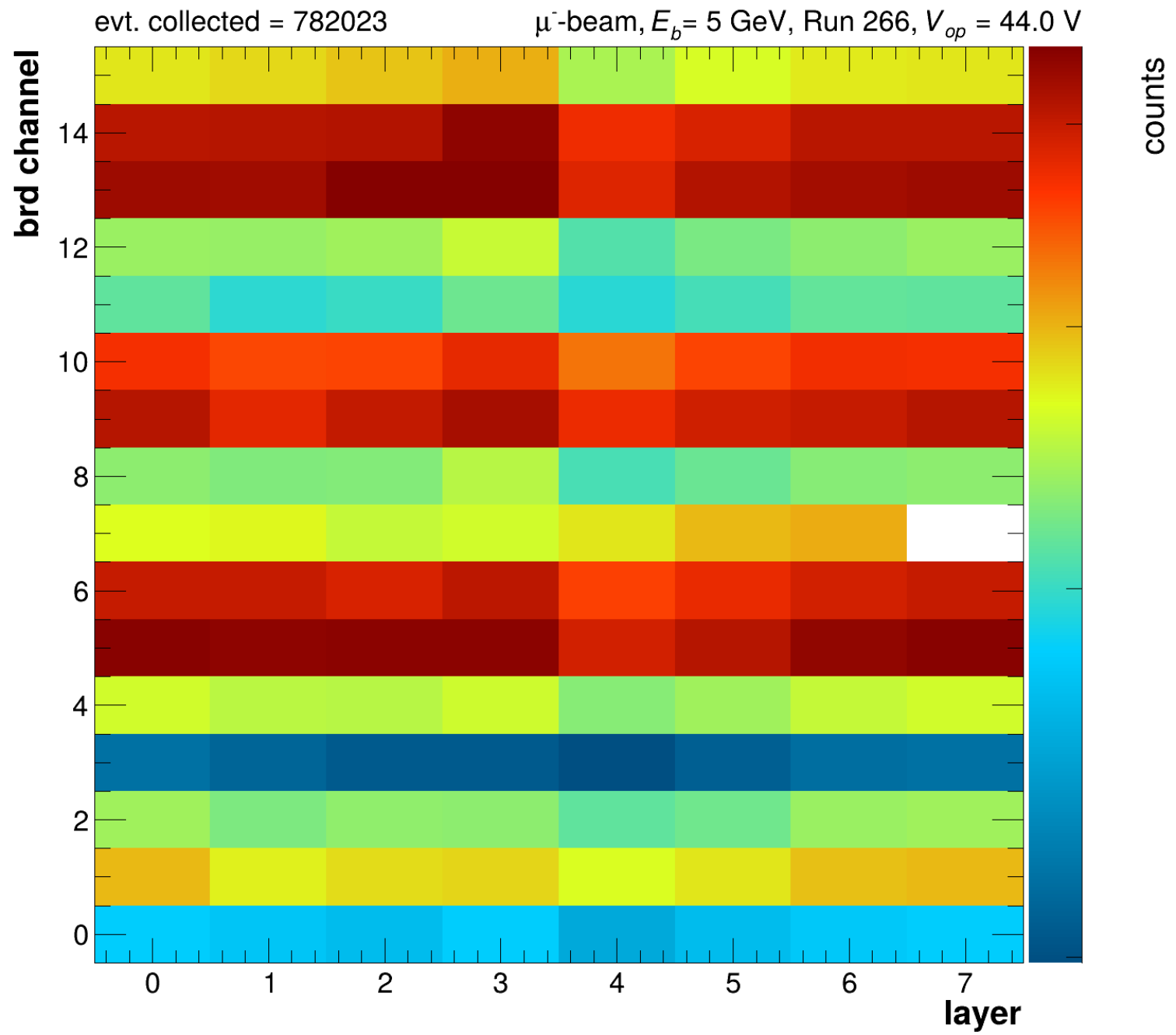

r:1 c:0 m:1
Total events: 517061
Landau-Gauss fit, trigg.
L MPV = 29.49 ± 0.07
Max = 31.34 ± 0.07
χ²/ndf = 15.12
SNR = 2.76
r:1 c:1 m:1
Total events: 515313
Landau-Gauss fit, trigg.
L MPV = 28.69 ± 0.05
Max = 30.43 ± 0.06
χ²/ndf = 20.19
SNR = 2.29
r:1 c:2 m:1
Total events: 513941
Landau-Gauss fit, trigg.
L MPV = 28.86 ± 0.06
Max = 31.06 ± 0.07
χ²/ndf = 23.34
SNR = 2.36
r:1 c:3 m:1 layer 4
Total events: 502764
Landau-Gauss fit, trigg.
L MPV = 33.26 ± 0.08
Max = 35.62 ± 0.09
χ²/ndf = 22.14
SNR = 2.32
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 505148
Landau-Gauss fit, trigg.
L MPV = 30.76 ± 0.09
Max = 33.11 ± 0.10
χ²/ndf = 17.44
SNR = 1.83
r:0 c:1 m:1
Total events: 516385
Landau-Gauss fit, trigg.
L MPV = 28.80 ± 0.06
Max = 30.62 ± 0.06
χ²/ndf = 20.17
SNR = 1.84
r:0 c:2 m:1
Total events: 501933
Landau-Gauss fit, trigg.
L MPV = 34.17 ± 0.09
Max = 36.86 ± 0.09
χ²/ndf = 26.56
SNR = 2.25
r:0 c:3 m:1
Total events: 492579
Landau-Gauss fit, trigg.
L MPV = 29.36 ± 0.08
Max = 31.32 ± 0.09
χ²/ndf = 11.94
SNR = 1.49
r:1 c:0 m:0
Total events: 781558
Landau-Gauss fit, trigg.
L MPV = 28.33 ± 0.06
Max = 29.96 ± 0.07
χ²/ndf = 16.60
SNR = 2.31
r:1 c:1 m:0
Total events: 781024
Landau-Gauss fit, trigg.
L MPV = 30.78 ± 0.06
Max = 32.77 ± 0.06
χ²/ndf = 25.06
SNR = 2.55
r:1 c:2 m:0
Total events: 781082
Landau-Gauss fit, trigg.
L MPV = 27.97 ± 0.06
Max = 30.03 ± 0.07
χ²/ndf = 20.52
SNR = 1.91
r:1 c:3 m:0
Total events: 779249
Landau-Gauss fit, trigg.
L MPV = 29.52 ± 0.07
Max = 31.24 ± 0.07
χ²/ndf = 23.72
SNR = 1.92
r:0 c:0 m:0
Total events: 779642
Landau-Gauss fit, trigg.
L MPV = 30.57 ± 0.08
Max = 32.19 ± 0.08
χ²/ndf = 15.22
SNR = 2.26
r:0 c:1 m:0
Total events: 781435
Landau-Gauss fit, trigg.
L MPV = 28.37 ± 0.06
Max = 30.09 ± 0.07
χ²/ndf = 15.74
SNR = 2.20
r:0 c:2 m:0
Total events: 779128
Landau-Gauss fit, trigg.
L MPV = 28.25 ± 0.06
Max = 29.78 ± 0.07
χ²/ndf = 16.65
SNR = 2.76
r:0 c:3 m:0
Total events: 777265
Landau-Gauss fit, trigg.
L MPV = 28.86 ± 0.09
Max = 30.71 ± 0.09
χ²/ndf = 7.41
SNR = 3.36
counts
ADC (arb. units)

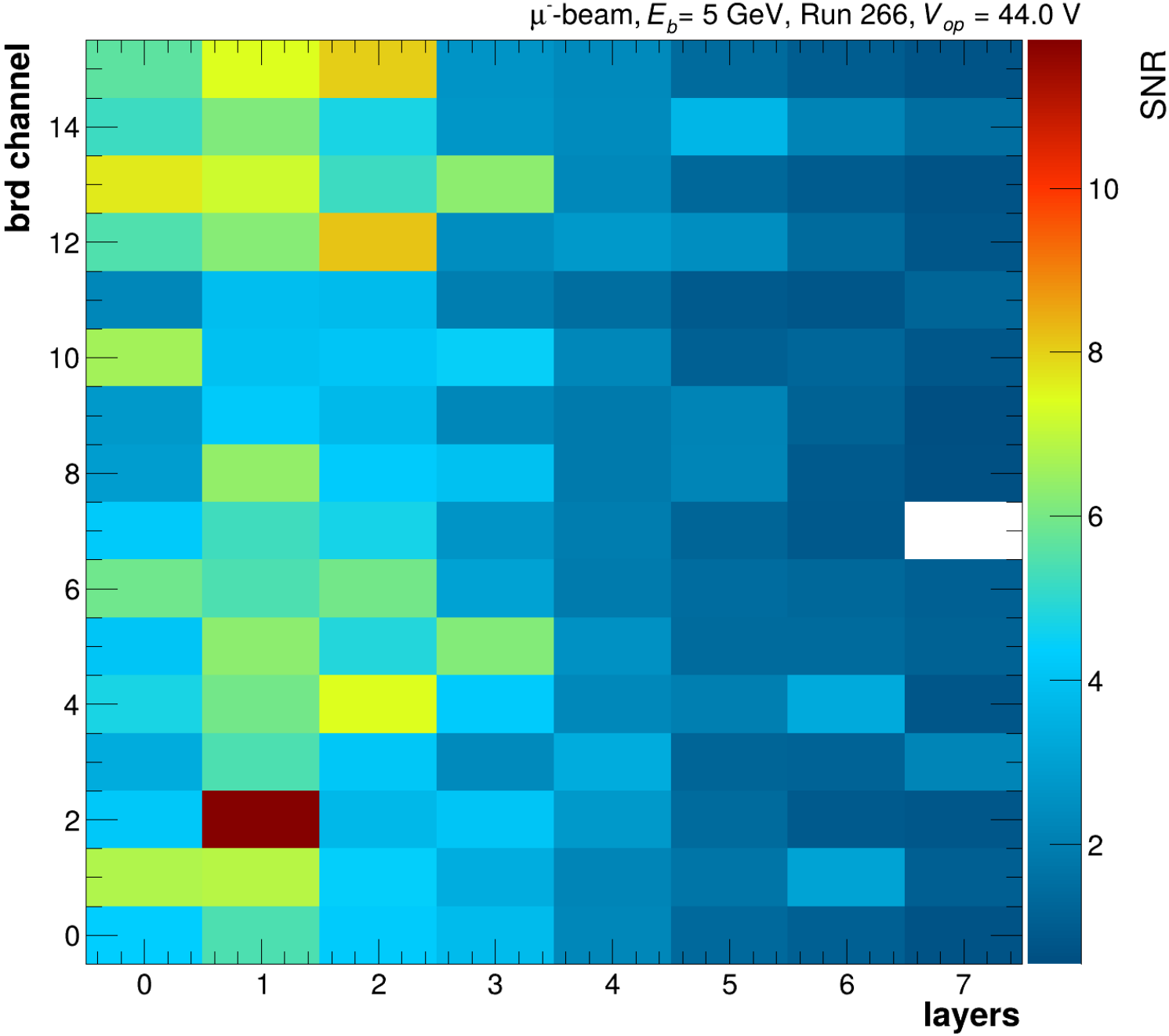

r:1 c:0 m:1
Total events: 508688
Landau-Gauss fit, trigg.
L MPV = 32.73 ± 0.08
Max = 34.85 ± 0.09
χ²/ndf = 18.69
SNR = 2.48
r:1 c:1 m:1
Total events: 509362
Landau-Gauss fit, trigg.
L MPV = 28.27 ± 0.06
Max = 30.03 ± 0.06
χ²/ndf = 28.42
SNR = 1.31
r:1 c:2 m:1
Total events: 452563
Landau-Gauss fit, trigg.
L MPV = 35.30 ± 0.08
Max = 37.81 ± 0.09
χ²/ndf = 27.60
SNR = 3.64
r:1 c:3 m:1 layer 5
Total events: 502498
Landau-Gauss fit, trigg.
L MPV = 31.31 ± 0.09
Max = 33.76 ± 0.10
χ²/ndf = 19.52
SNR = 1.42
μ⁻-beam, E_b = 5 GeV
Run 266, V_{op} = 44.0 V
r:0 c:0 m:1
Total events: 517661
Landau-Gauss fit, trigg.
L MPV = 34.63 ± 0.12
Max = 37.28 ± 0.13
χ²/ndf = 24.78
SNR = 2.21
r:0 c:1 m:1
Total events: 506874
Landau-Gauss fit, trigg.
L MPV = 33.18 ± 0.07
Max = 35.35 ± 0.08
χ²/ndf = 30.47
SNR = 2.14
r:0 c:2 m:1
Total events: 500531
Landau-Gauss fit, trigg.
L MPV = 35.73 ± 0.10
Max = 38.54 ± 0.11
χ²/ndf = 22.82
SNR = 1.09
r:0 c:3 m:1
Total events: 513859
Landau-Gauss fit, trigg.
L MPV = 34.03 ± 0.14
Max = 36.85 ± 0.15
χ²/ndf = 24.06
SNR = 0.92
r:1 c:0 m:0
Total events: 779952
Landau-Gauss fit, trigg.
L MPV = 35.07 ± 0.09
Max = 37.36 ± 0.10
χ²/ndf = 17.87
SNR = 2.03
r:1 c:1 m:0
Total events: 779990
Landau-Gauss fit, trigg.
L MPV = 35.31 ± 0.07
Max = 37.65 ± 0.08
χ²/ndf = 21.88
SNR = 1.39
r:1 c:2 m:0
Total events: 755387
Landau-Gauss fit, trigg.
L MPV = 28.38 ± 0.06
Max = 30.29 ± 0.07
χ²/ndf = 16.46
SNR = 1.45
r:1 c:3 m:0
Total events: 779337
Landau-Gauss fit, trigg.
L MPV = 31.42 ± 0.08
Max = 33.63 ± 0.09
χ²/ndf = 19.48
SNR = 1.21
r:0 c:0 m:0
Total events: 781877
Landau-Gauss fit, trigg.
L MPV = 31.67 ± 0.09
Max = 33.76 ± 0.10
χ²/ndf = 16.53
SNR = 1.35
r:0 c:1 m:0
Total events: 779904
Landau-Gauss fit, trigg.
L MPV = 33.03 ± 0.08
Max = 35.25 ± 0.09
χ²/ndf = 30.32
SNR = 1.70
r:0 c:2 m:0
Total events: 779211
Landau-Gauss fit, trigg.
L MPV = 33.82 ± 0.09
Max = 36.15 ± 0.10
χ²/ndf = 17.15
SNR = 1.40
r:0 c:3 m:0
Total events: 781325
Landau-Gauss fit, trigg.
L MPV = 30.98 ± 0.09
Max = 32.74 ± 0.10
χ²/ndf = 7.60
SNR = 1.23
counts
ADC (arb. units)

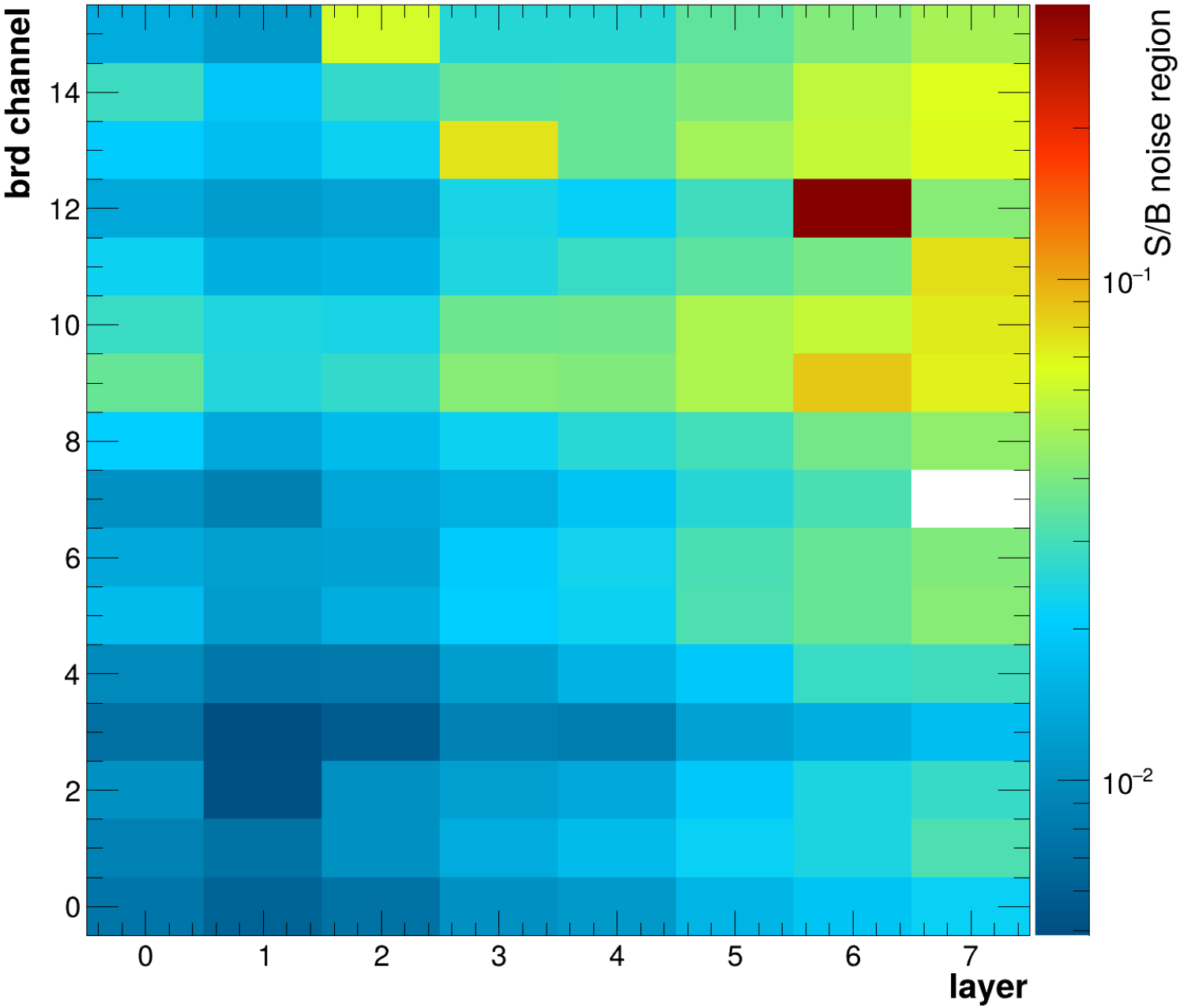

r:1 c:0 m:1
Total events: 80133
Landau-Gauss fit, trigg.
L MPV = 41.49 ± 0.15
Max = 44.74 ± 0.16
χ²/ndf = 24.13
SNR = 1.39
r:1 c:1 m:1
Total events: 511930
Landau-Gauss fit, trigg.
L MPV = 38.73 ± 0.10
Max = 41.61 ± 0.11
χ²/ndf = 19.14
SNR = 0.94
r:1 c:2 m:1
Total events: 505455
Landau-Gauss fit, trigg.
L MPV = 33.57 ± 0.09
Max = 35.89 ± 0.09
χ²/ndf = 32.30
SNR = 2.17
r:1 c:3 m:1 layer 6
Total events: 512447
Landau-Gauss fit, trigg.
L MPV = 38.75 ± 0.13
Max = 41.69 ± 0.14
χ²/ndf = 18.63
SNR = 1.01
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 517010
Landau-Gauss fit, trigg.
L MPV = 37.86 ± 0.13
Max = 40.67 ± 0.14
χ²/ndf = 12.70
SNR = 0.91
r:0 c:1 m:1
Total events: 366905
Landau-Gauss fit, trigg.
L MPV = 43.19 ± 0.14
Max = 46.61 ± 0.16
χ²/ndf = 35.11
SNR = 1.15
r:0 c:2 m:1
Total events: 509164
Landau-Gauss fit, trigg.
L MPV = 36.10 ± 0.11
Max = 38.69 ± 0.11
χ²/ndf = 20.51
SNR = 1.26
r:0 c:3 m:1
Total events: 512342
Landau-Gauss fit, trigg.
L MPV = 39.63 ± 0.16
Max = 42.63 ± 0.17
χ²/ndf = 11.30
SNR = 0.78
r:1 c:0 m:0
Total events: 559972
Landau-Gauss fit, trigg.
L MPV = 35.21 ± 0.10
Max = 37.75 ± 0.11
χ²/ndf = 21.83
SNR = 3.29
r:1 c:1 m:0
Total events: 781065
Landau-Gauss fit, trigg.
L MPV = 37.31 ± 0.09
Max = 39.99 ± 0.10
χ²/ndf = 23.74
SNR = 1.40
r:1 c:2 m:0
Total events: 779785
Landau-Gauss fit, trigg.
L MPV = 36.99 ± 0.09
Max = 39.60 ± 0.10
χ²/ndf = 34.21
SNR = 1.30
r:1 c:3 m:0
Total events: 781218
Landau-Gauss fit, trigg.
L MPV = 37.78 ± 0.13
Max = 40.79 ± 0.14
χ²/ndf = 13.40
SNR = 0.88
r:0 c:0 m:0
Total events: 781781
Landau-Gauss fit, trigg.
L MPV = 39.50 ± 0.15
Max = 42.57 ± 0.17
χ²/ndf = 13.09
SNR = 1.09
r:0 c:1 m:0
Total events: 707495
Landau-Gauss fit, trigg.
L MPV = 33.70 ± 0.10
Max = 36.13 ± 0.11
χ²/ndf = 27.85
SNR = 3.08
r:0 c:2 m:0
Total events: 780268
Landau-Gauss fit, trigg.
L MPV = 34.95 ± 0.14
Max = 37.74 ± 0.15
χ²/ndf = 19.45
SNR = 0.90
r:0 c:3 m:0
Total events: 781212
Landau-Gauss fit, trigg.
L MPV = 35.57 ± 0.14
Max = 38.11 ± 0.15
χ²/ndf = 7.76
SNR = 1.18
counts
ADC (arb. units)

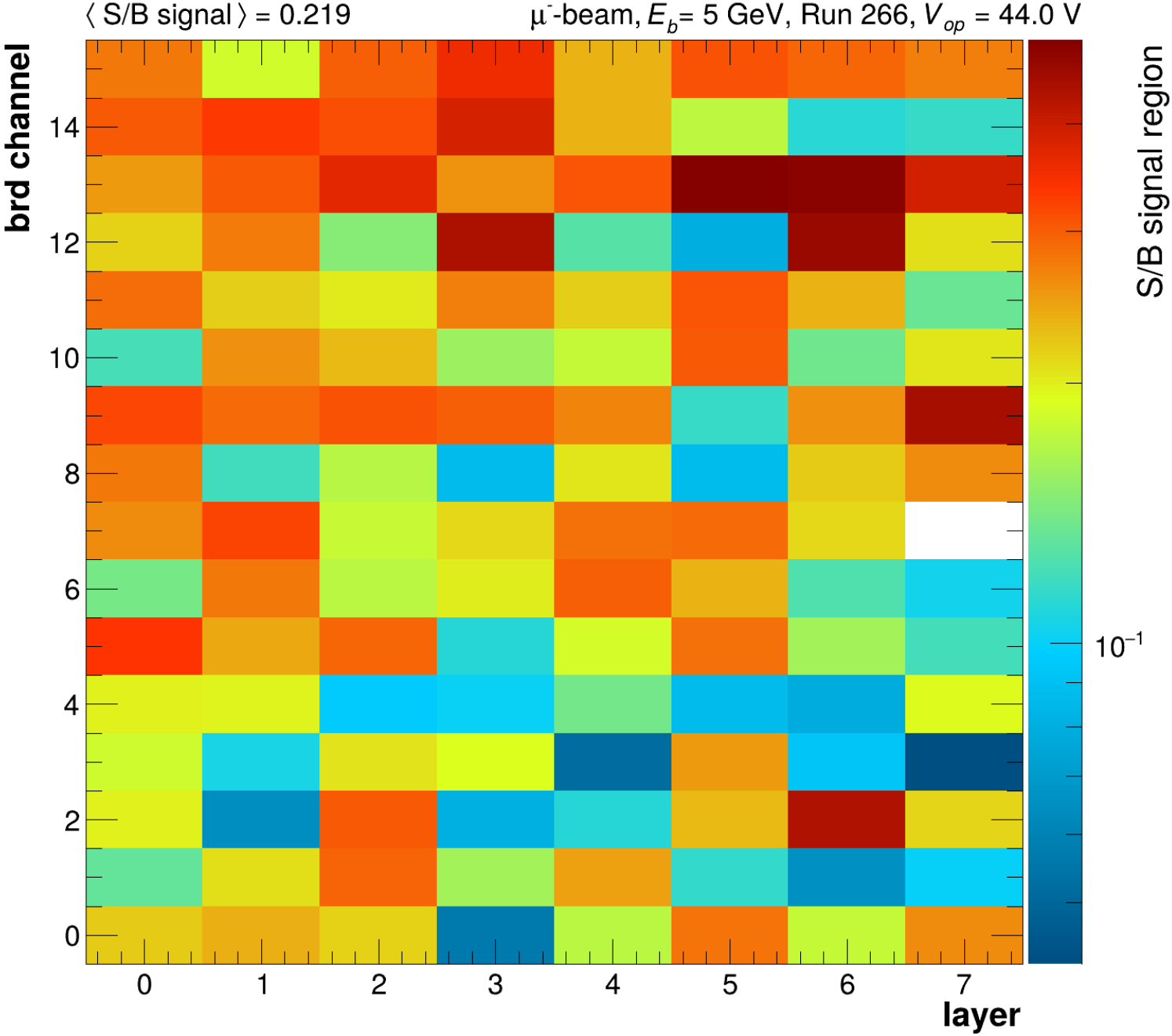

r:1 c:0 m:1
Total events: 517221
Landau-Gauss fit, trigg.
L MPV = 42.15 ± 0.18
Max = 45.52 ± 0.20
χ²/ndf = 18.11
SNR = 0.77
r:1 c:1 m:1
Total events: 508884
Landau-Gauss fit, trigg.
L MPV = 42.42 ± 0.14
Max = 45.66 ± 0.15
χ²/ndf = 26.48
SNR = 0.71
r:1 c:2 m:1
Total events: 505585
Landau-Gauss fit, trigg.
L MPV = 39.75 ± 0.13
Max = 42.85 ± 0.14
χ²/ndf = 32.99
SNR = 1.50
r:1 c:3 m:1 layer 7
Total events: 499327
Landau-Gauss fit, trigg.
L MPV = 41.55 ± 0.17
Max = 44.83 ± 0.19
χ²/ndf = 19.97
SNR = 0.72
μ⁻-beam, E_b = 5 GeV
Run 266, V_op = 44.0 V
r:0 c:0 m:1
Total events: 513501
Landau-Gauss fit, trigg.
L MPV = 40.56 ± 0.18
Max = 43.74 ± 0.19
χ²/ndf = 12.53
SNR = 0.61
r:0 c:1 m:1
Total events: 508573
Landau-Gauss fit, trigg.
L MPV = 42.70 ± 0.15
Max = 46.00 ± 0.16
χ²/ndf = 26.45
SNR = 0.54
r:0 c:2 m:1
Total events: 452296
Landau-Gauss fit, trigg.
L MPV = 42.95 ± 0.18
Max = 46.34 ± 0.20
χ²/ndf = 22.44
SNR = 0.84
r:0 c:3 m:1
Total events: 297412
Landau-Gauss fit, trigg.
L MPV = 44.52 ± 0.23
Max = 48.14 ± 0.24
χ²/ndf = 18.18
SNR = 1.23
r:1 c:0 m:0
Total events: 781911
Landau-Gauss fit, trigg.
L MPV = 40.59 ± 0.17
Max = 43.68 ± 0.18
χ²/ndf = 17.05
SNR = 0.77
r:1 c:1 m:0
Total events: 780256
Landau-Gauss fit, trigg.
L MPV = 42.43 ± 0.14
Max = 45.79 ± 0.15
χ²/ndf = 29.70
SNR = 1.13
r:1 c:2 m:0
Total events: 779948
Landau-Gauss fit, trigg.
L MPV = 42.15 ± 0.14
Max = 45.44 ± 0.16
χ²/ndf = 32.90
SNR = 1.08
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 781415
Landau-Gauss fit, trigg.
L MPV = 40.15 ± 0.19
Max = 43.27 ± 0.20
χ²/ndf = 8.63
SNR = 0.72
r:0 c:1 m:0
Total events: 780110
Landau-Gauss fit, trigg.
L MPV = 41.40 ± 0.16
Max = 44.63 ± 0.18
χ²/ndf = 16.78
SNR = 1.04
r:0 c:2 m:0
Total events: 755384
Landau-Gauss fit, trigg.
L MPV = 42.24 ± 0.21
Max = 45.57 ± 0.22
χ²/ndf = 34.77
SNR = 0.81
r:0 c:3 m:0
Total events: 670645
Landau-Gauss fit, trigg.
L MPV = 43.56 ± 0.24
Max = 47.17 ± 0.26
χ²/ndf = 14.09
SNR = 2.21
counts
ADC (arb. units)

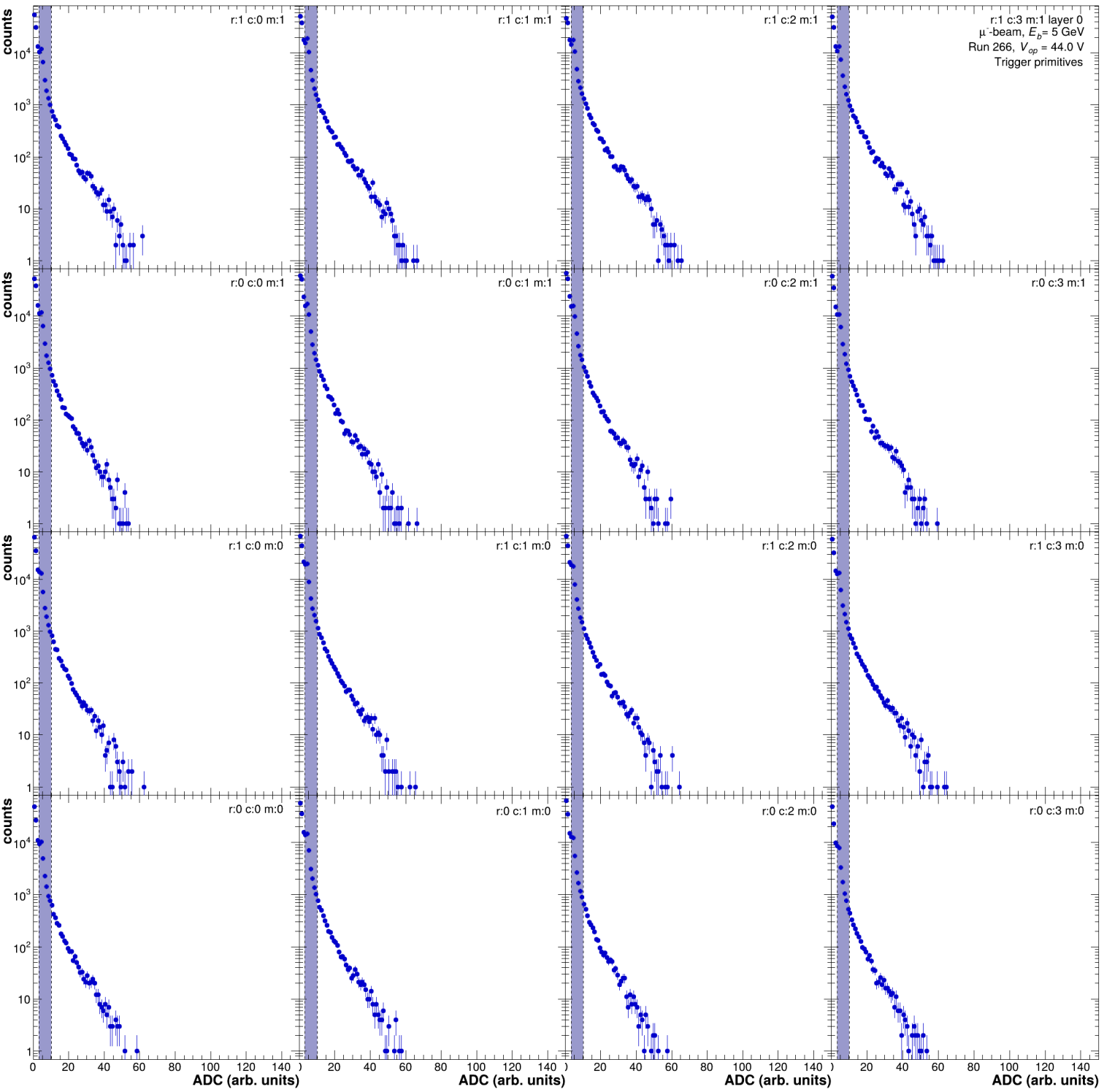

μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
Y (row)
X (col)
Num triggers
×10³
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
400
350
300
250
200

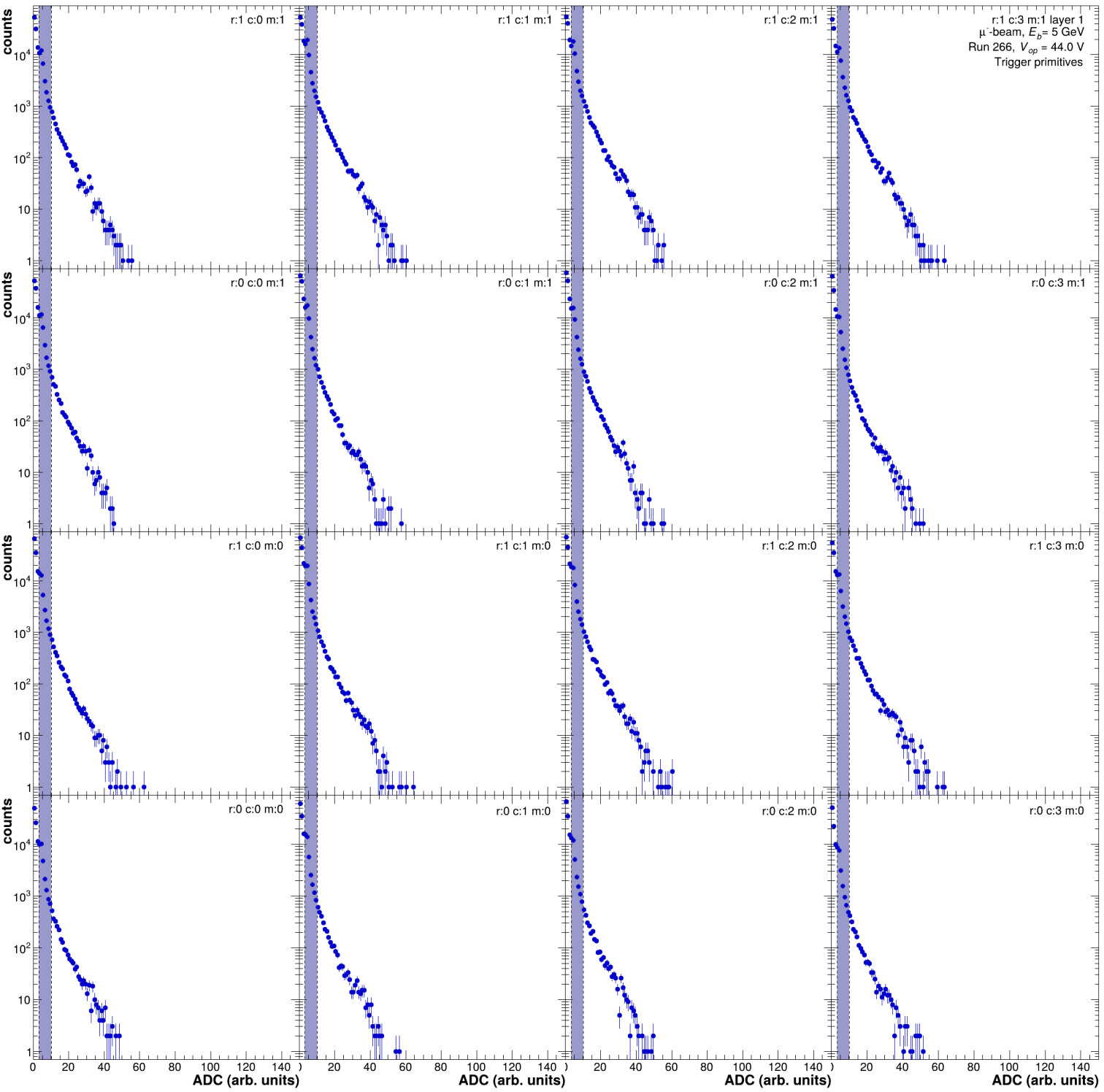

evt. collected = 782023
μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
counts

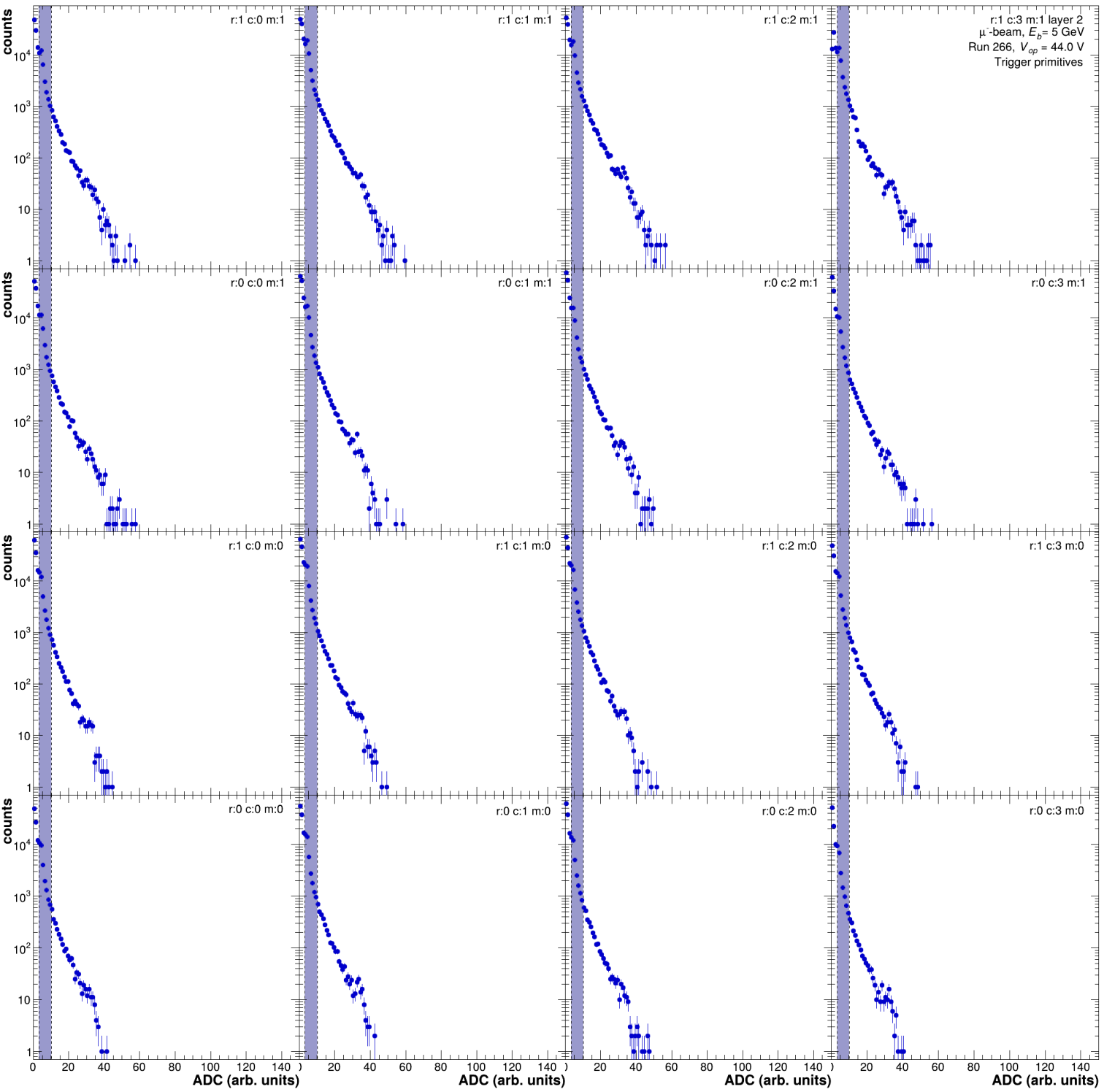

μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
layers
0
1
2
3
4
5
6
7
SNR
2
4
6
8
10

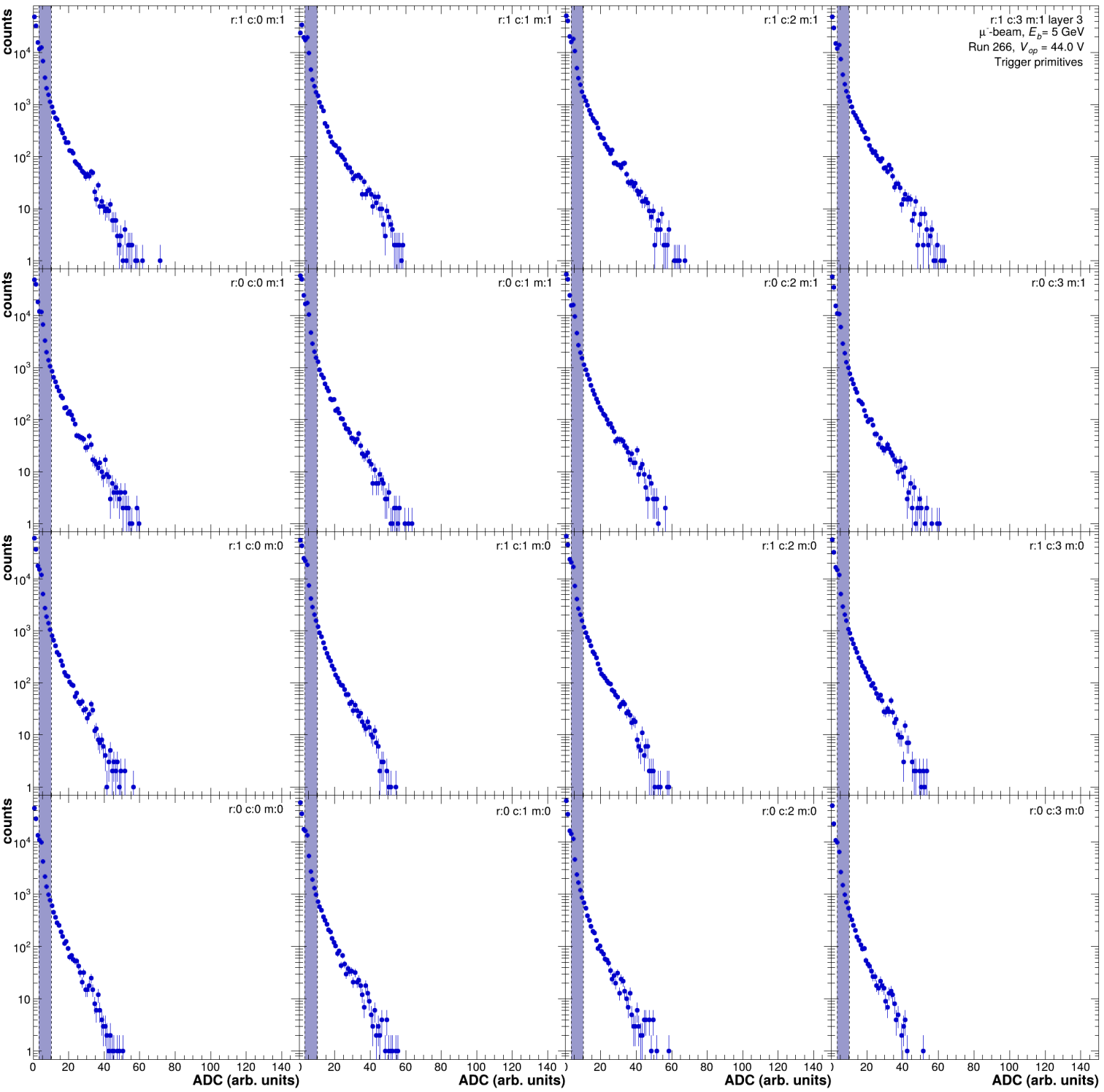

⟨ S/B noise ⟩ = 0.029
μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B noise region
10^{-1}
10^{-2}

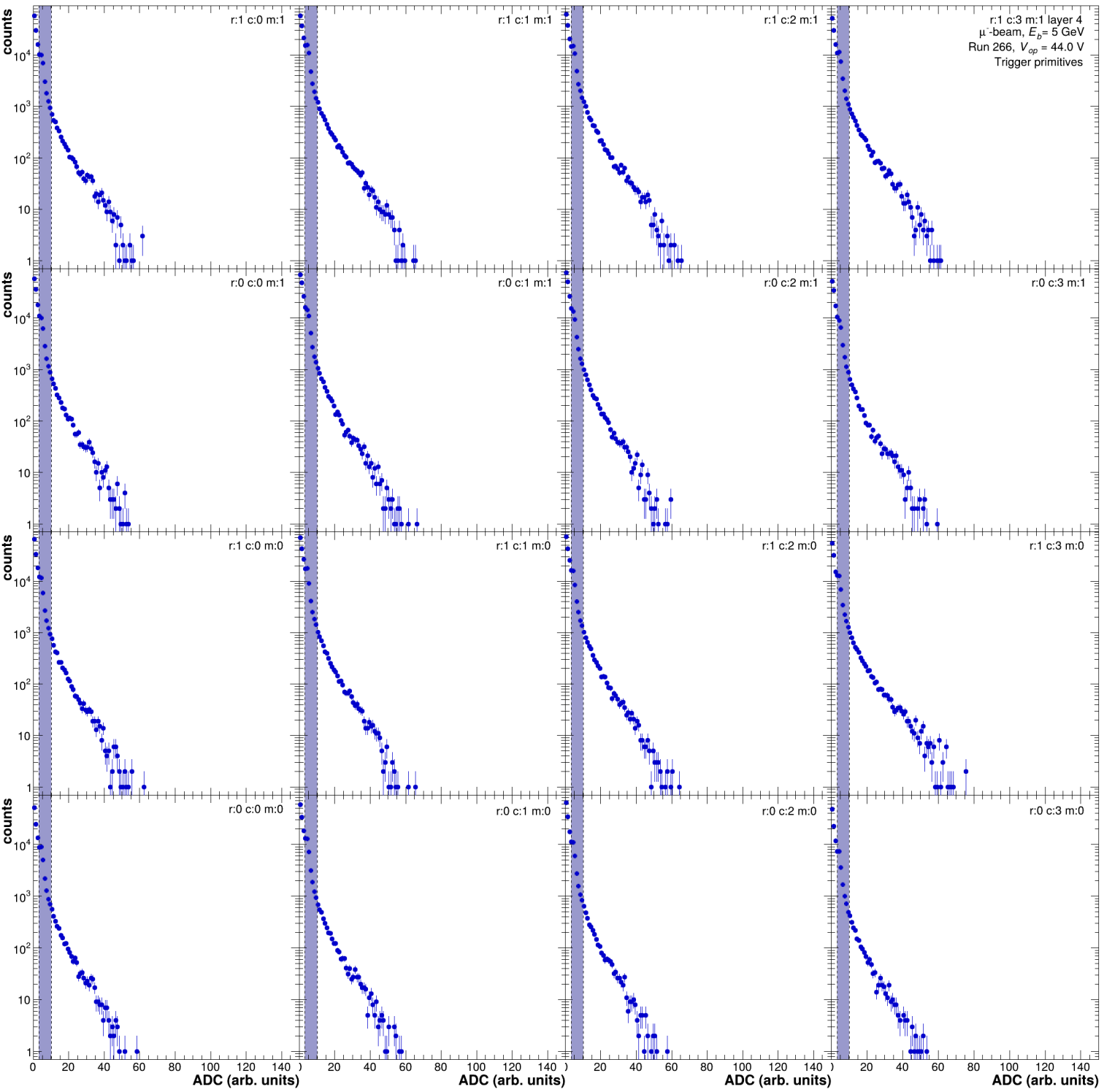

⟨ S/B signal ⟩ = 0.219
μ⁻-beam, E_b= 5 GeV, Run 266, V_{op} = 44.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region
10^{-1}

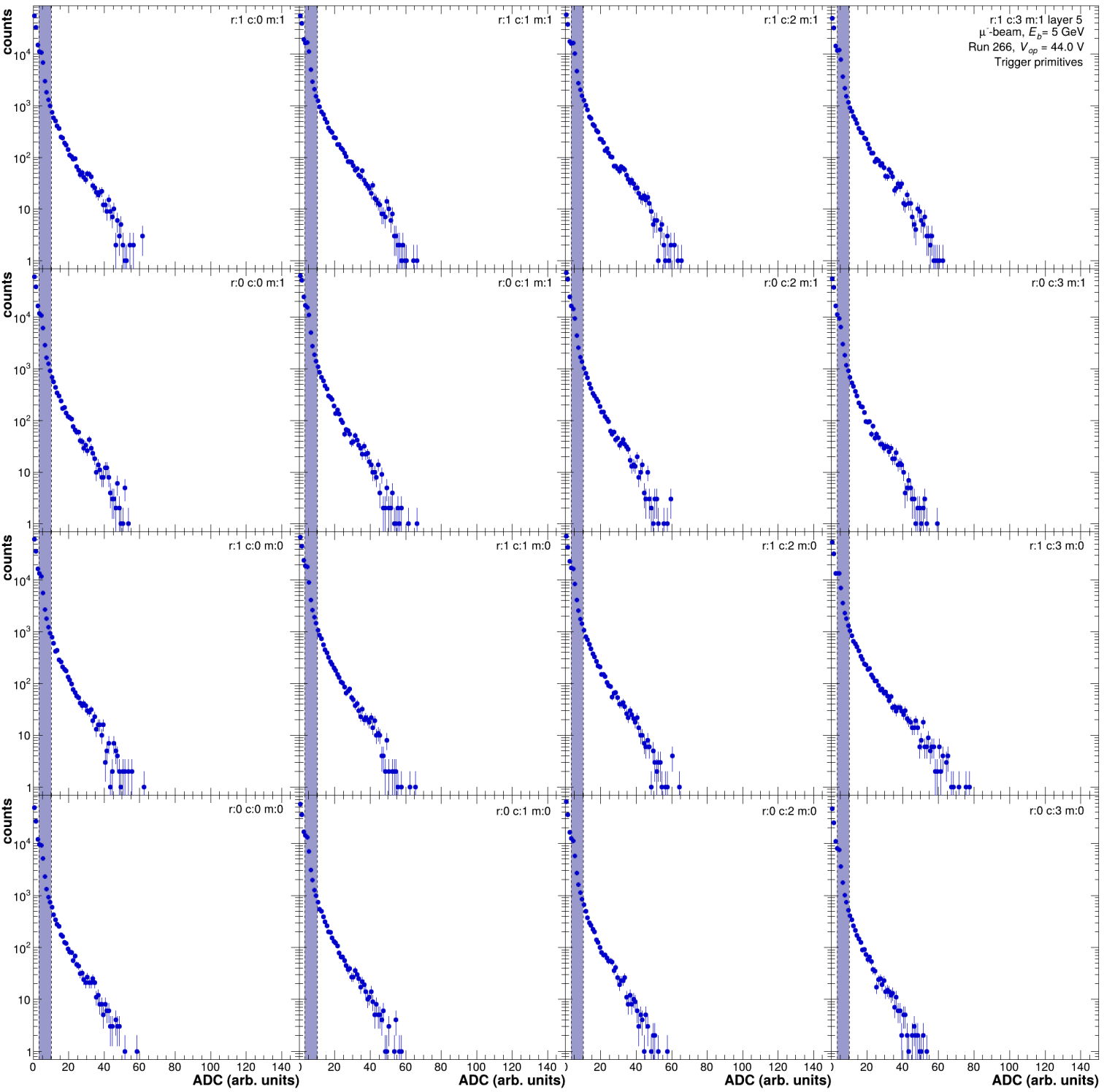

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

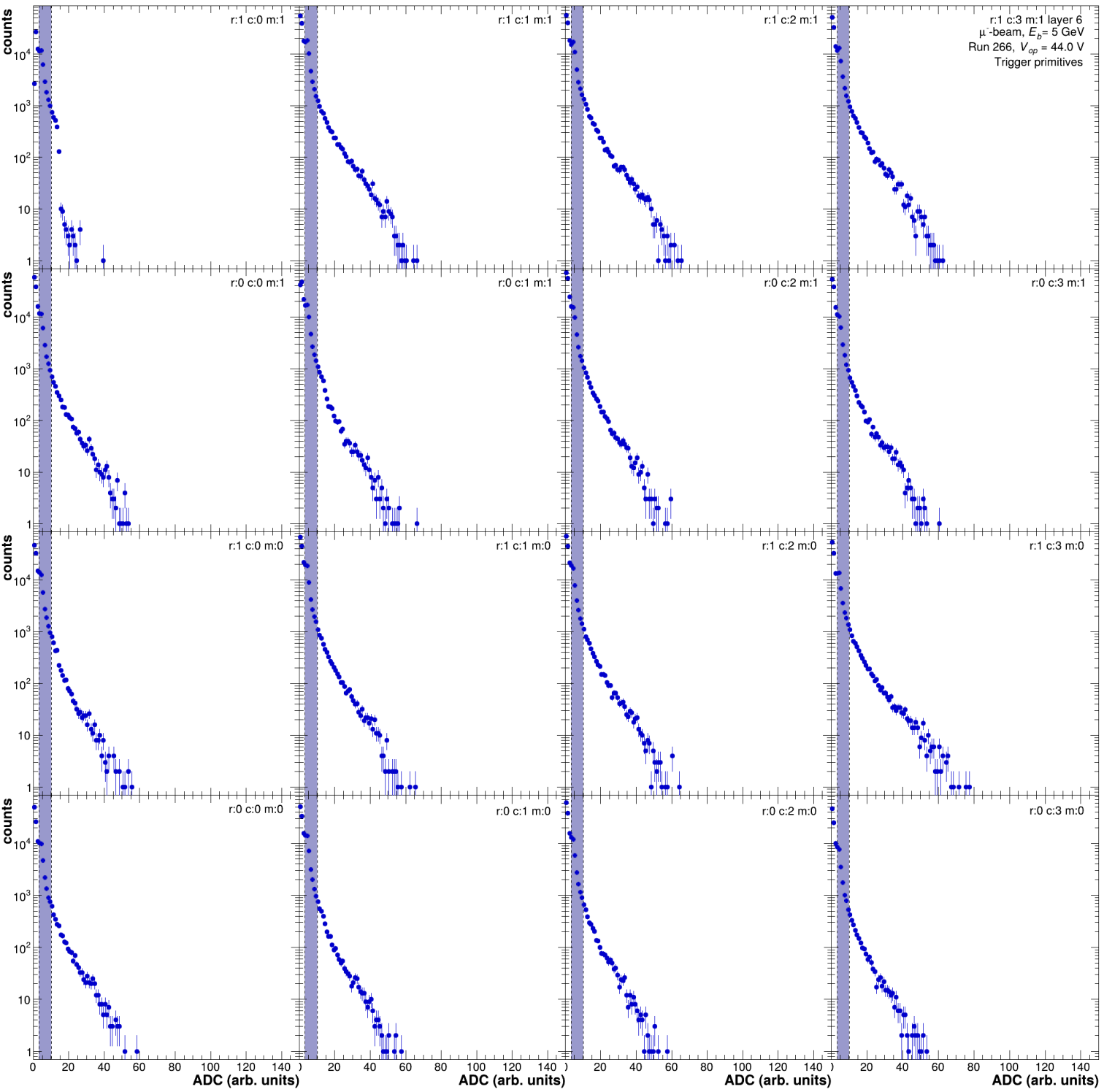

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

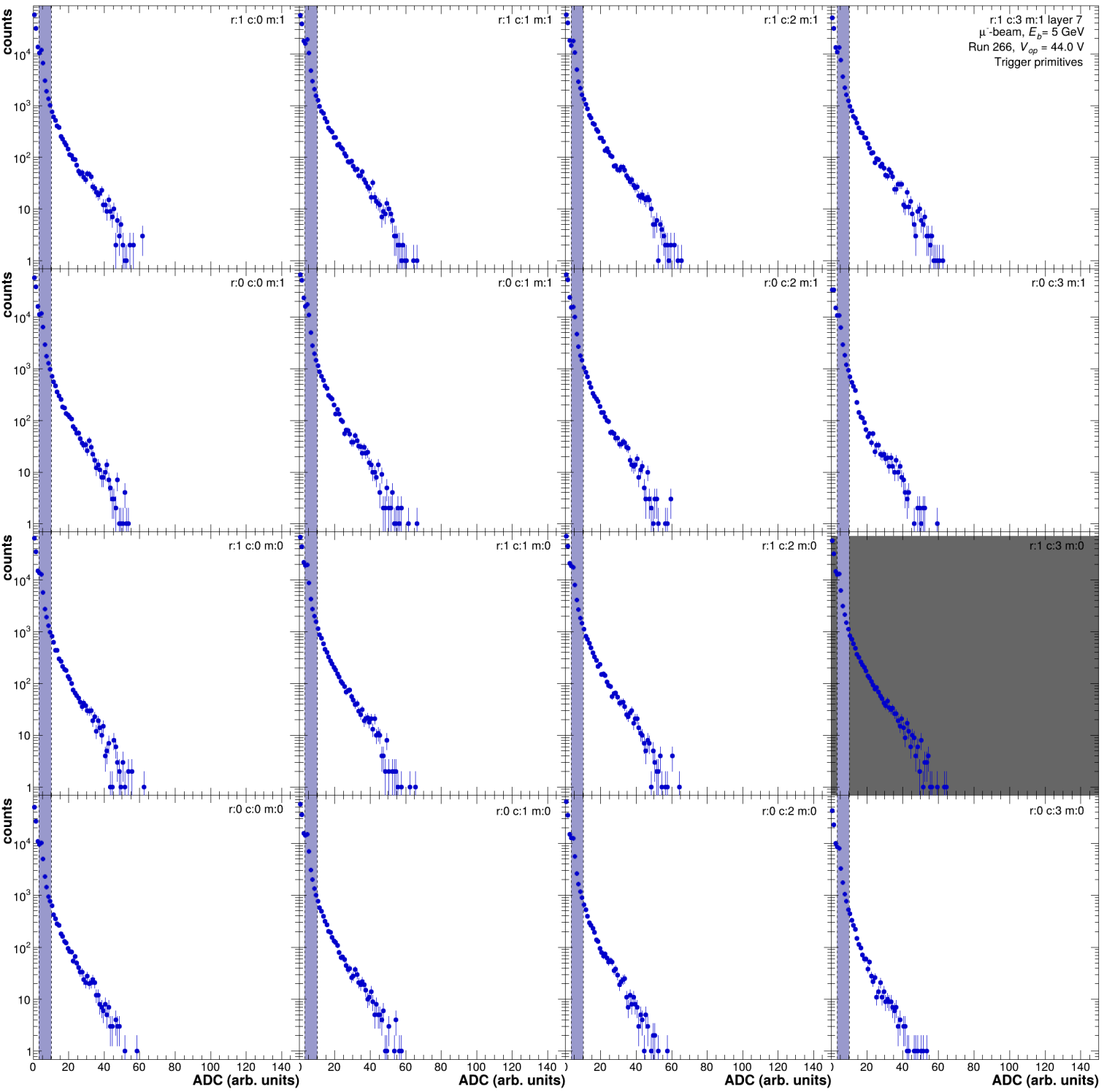

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
μ⁻-beam, E_b= 5 GeV
Run 266, V_{op} = 44.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)